

Chapter VIII

Parameter Spaces and Mandelbrot Sets

1 The Idea of a Parameter Space: A Map of Fractals

A map with nothing marked on it is practically useless. A map of a 1000×1000 square mile region containing the British Isles is shown in Figure VIII.217; it does not convey much information. However, as a concept it is quite exciting. Each location on the map corresponds to somewhere on Earth. For example, the dot with coordinates (750, 227.3) represents the town of Maidstone. A point on the map may represent a certain grain of soil in a ploughed field, or a molecule of flotsam on the top of some foam on the surface of the sea. Nearby points in the map correspond to nearby points on the Earth. Connected sets with interiors correspond to physical regions.

How could a perfect map be made? Ideally, it should specify locations on the Earth's surface at a certain instant. The coordinates would be relative to some absolute coordinate system, perhaps determined by reference to the fixed stars. Moreover, the surface of the Earth would have to be defined precisely, up to the last molecule of water, soil, and plant matter; for this purpose one can imagine using a straight line from the center of the Earth as suggested in Figure VIII.218. Of course, maps are not made like this, but the goal is the same: to have an accurate correspondence between points on the physical surface of the Earth and the physical surface of the paper.

We must be careful how we interpret a map. Geographical maps are complicated by the real number system and the unphysical notion of infinite divisibility. Mathematically, the map is an abstract place. A point on the map cannot represent a certain physical atom in the real world, not just because of inaccuracies in the map, but because of the dual nature of matter: according to current theories one cannot know the exact location of an atom at a given instant.

Fractal geometers avoid this problem by pretending that the surface of the Earth is an abstract place too; we imagine, once again, that matter is infinitely divisible, and

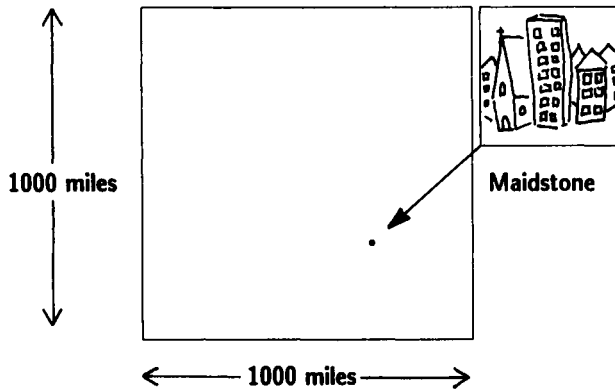


Figure VIII.217. A map with no information marked on it is quite exciting. A very unhelpful map of the British Isles. The dot represents Maidstone. The map is accurate up to dot size.

that we can address every point. In the same spirit, we presume that we can model trees, clouds, horizons, churning seas, and infinitely finely defined coastlines. Then, for example, we can define the Hausdorff-Besicovitch dimension of the coastline of the British Isles.

For a map to be useful it must have information marked on it, such as heights above sea level, population densities, roads, vegetation, rainfall, types of underlying rock, ownership, names, incidence of volcanoes, malarial infestation, and so on. A good way of providing such information is with colors. For example, if we use blue for water and green for land then we can “see” the land on the map and we can understand some geometrical relationships. We can estimate overland distances between points, land areas of islands, the shortest sea passage from Llanelli Bay

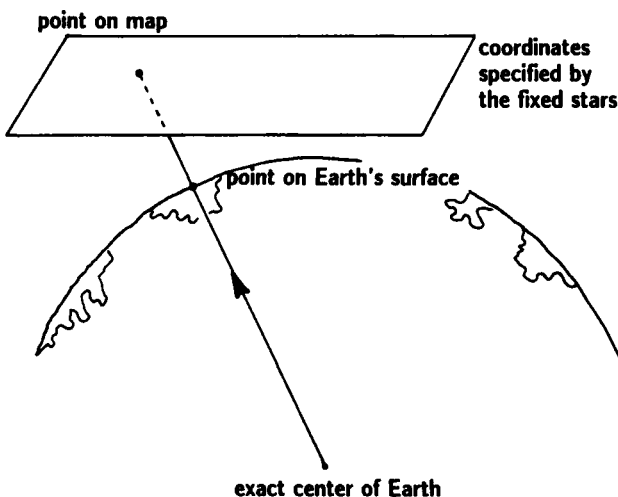
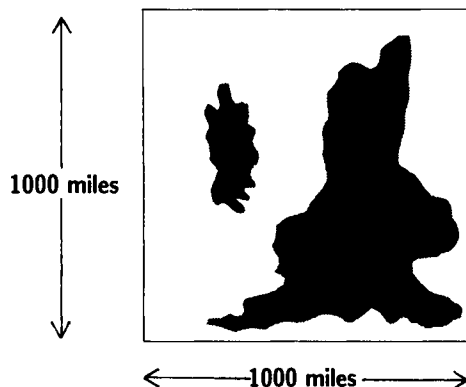


Figure VIII.218. How Figure VIII.217 might have been made.

Figure VIII.219. In this map points corresponding to land have been shaded. A fascinating entity, the coastline, is revealed.



to Amylwhc Harbour, the length of the coastline, etc. All this is achieved through the device of marking some colors on a blank map!

Let us consider the boundary of the shaded region in Figure VIII.219. It is here that the map conveys extra information. In the interior of the shaded region we learn no more about the surface of the Earth than that “there is land there.” However, on the boundary we learn not only that “there are land and sea there,” but also, if the map is accurate enough, a feature that we will actually “see” on the surface, namely the local shape of the coastline.

The latter idea can be extended. If we include more colors on a geographical map, to provide more information about properties of the Earth’s surface, we produce more boundaries on the map. These boundaries can give information about local geometry. For example, a map finely colored according to elevation reveals the shapes of the bases of the mountains, the paths of rivers, and – if we look closely enough—the outlines of buildings. Such a map, made abstract and perfect, placed in a metric space, would contain much detailed information about what, at each point, the local observer would see.

Examples & Exercises

1.1. Study an atlas that contains maps colored according to diverse criteria, such as rainfall, population density, vegetation, and elevation. Discuss to what extent these maps provide information about the local geometry of the surface of the Earth.

We turn attention to making colored maps of parameterized families of fractals. We consider families of iterated function systems and families of dynamical systems that depend on two real parameters. The collection of possible parameter values defines a *parameter space* associated with the family. We use the notation P to denote a parameter space. Typically P is a subspace of $(\mathbb{R}^2, \text{Euclidean})$ such as $\blacksquare$, a closed ball, or $\mathbb{R}^2$.

An example of a parameter space is $P = \{(\lambda_1, \lambda_2) \in \mathbb{R}^2 : |\lambda_1|, |\lambda_2| < 2^{-0.5}\}$. This

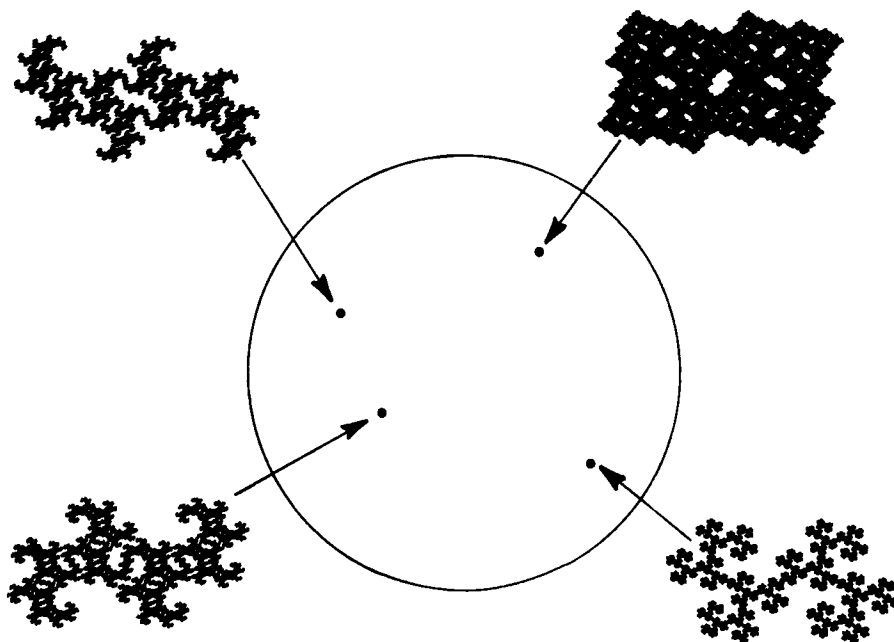


Figure VIII.220. An example of a parameter space. Each point λ in the space corresponds to a fractal, $A(\lambda)$. This is a poor map, because very little has been marked on it. It is like the map of the British Isles shown in Figure VIII.217. It needs coloring.

is a parameter space for the family of hyperbolic IFS $\{\mathbb{C}; (\lambda_1 + i\lambda_2)z + 1, (\lambda_1 + i\lambda_2)z - 1\}$. Each point $\lambda = (\lambda_1, \lambda_2) \in P$ corresponds to an IFS. Each IFS possesses a unique attractor, say $A(\lambda)$. Hence each point of P corresponds to a single fractal. We can think of P as representing part of $\mathcal{H}(\mathbb{C})$, a space of fractals. A map of P , with a few points marked on it, is shown in Figure VIII.220. Each point in P corresponds to a single fractal. Nearby points in P correspond to nearby fractals, that is, points in $\mathcal{H}(\mathbb{C})$ whose Hausdorff distance apart is small.

Another example is $P = \mathbb{C}$, which provides a parameter space for the family of dynamical systems $\{\hat{\mathbb{C}}; f_\lambda(z) = z^2 - \lambda\}$. Each point in the parameter space corresponds to a different dynamical system. Each dynamical system is associated with a unique Julia set, $J(\lambda)$. The collection of fractals $\{J(\lambda) : \lambda \in P\}$, associated with the parameter space, is vast and diverse.

Let X be a two-dimensional metric space such as $\mathbb{R}^2$ or $\hat{\mathbb{C}}$. Let P denote a parameter space corresponding to a family of fractals $\{A(\lambda) \in \mathcal{H}(X) : \lambda \in P\}$. Can we provide the explorer of fractals, who wishes to investigate this collection of fractals, with a colored map? This map should give her information about the sets $A(\lambda)$ to be found at different points on P .

Suppose $P = \blacksquare$. To make a map, let us represent P by the set of pixels on the screen of a computer graphics monitor. The idea is to color the pixel λ according to some property of $A(\lambda)$. (We write $\lambda = \text{pixel} = \text{point in } P$, without repeatedly explaining that λ is a point in the small rectangle in P that corresponds to the pixel.)

Suitable properties could relate to the connectivity of $A(\lambda)$, the fractal dimension of $A(\lambda)$, the escape time near a special point on $A(\lambda)$ under an associated dynamical system, the number of holes in $A(\lambda)$, or the presence of straight lines in $A(\lambda)$, for example.

If we make a good selection of the properties to associate with colors, the result will be a useful map containing various differently colored regions. This map will be a ready reference for the explorer of fractals. It will tell him something about what to expect at he travels about P . He might be suprised nonetheless.

The boundaries of the colored regions can provide additional geometrical information to the explorer, over and above the information that the map was originally designed to convey. It sometimes occurs that the local shapes of the boundaries in the map reflect the shapes of the corresponding fractals. There is a deep principle here, which we shall not pin down as a theorem, but which we will illustrate in a number of cases.

Examples & Exercises

1.2. It is often useful, when establishing the Lipshitz continuity condition in Theorem 11.1 in Chapter III, to remove dependencies on $x \in X$ by restricting the domain of the IFS to a compact set B that contains the attractors $A(\lambda)$ for all values of $\lambda \in P$. Since this does not change the attractors themselves, continuous dependency on λ in B then guarantees it for the original IFS. This can be done in practice by means of the following lemma:

Lemma 1.1 *Let $B \in \mathcal{H}(X)$ such that for all $p \in P$, $W(p, B) \subset B$. Then for all $p \in P$, $A(p) \subset B$.*

Prove the lemma.

1.3. Let $P = \{(\lambda_1, \lambda_2) \in \mathbb{R}^2 : |\lambda_1|, |\lambda_2| \leq 0.9\}$. The family of IFS $\{\mathbb{R}; \lambda_1 x, \lambda_2 x + 1 - \lambda_2, 0.5x + 0.5\}$ is hyperbolic, with contractivity factor $s = 0.9$, for all $\lambda \in P$. Use Theorem 3.11.1 and the lemma of the previous problem to prove that the attractor $A(\lambda)$ depends continuously on λ .

1.4. The family of IFS $\{[0, 1]; \lambda_1 x^2, \lambda_2 x + (1 - \lambda_2)\}$ is hyperbolic, with contractivity factor $s = 0.9$, for all λ in the parameter space

$$P = \{(\lambda_1, \lambda_2) \in \mathbb{R}^2 : 0 \leq \lambda_1 \leq 0.45, 0 \leq \lambda_2 \leq 0.9\}.$$

Since we have

$$\begin{aligned} d(w_{1_p}(x), w_{1_q}(x)) &= |x^2| |\lambda_{1_p} - \lambda_{1_q}| \leq |\lambda_{1_p} - \lambda_{1_q}| \\ d(w_{2_p}(x), w_{2_q}(x)) &= |x - 1| |\lambda_{2_p} - \lambda_{2_q}| \leq |\lambda_{2_p} - \lambda_{2_q}|, \end{aligned}$$

we can satisfy the Lipshitz condition with $k = 1$ in both cases. The attractor depends continuously on λ .

1.5. Sometimes the Lipshitz condition is not that convenient to prove directly. In this case, we can go directly to the continuity statement that it is used to guarantee.

Suppose we can show directly that $w_i(p, x)$ is a continuous function from $P \times \mathbf{X} \rightarrow \mathbf{X}$. That is, *without holding either p or x fixed*, given $\epsilon > 0$, there is a $\delta_i > 0$ for each $w_i(p, x)$, such that

$$d(w_i(p, x), w_i(q, y)) < \epsilon \quad \text{whenever} \quad d((p, x), (q, y)) < \delta_i.$$

Then the conclusions of Theorem 11.1 in Chapter III hold. This does not mean that the Lipschitz condition (or something like it) is unnecessary; such conditions guarantee the statement we have just made. If we can guarantee it some other way, it is equivalent.

1.6. An example of a parameter space is $(P, \text{Euclidean})$, where $P = \{(\lambda_1, \lambda_2) \in \mathbb{R}^2 : |\lambda_1|, |\lambda_2| \leq 0.999\}$. This space can be used to represent the family of hyperbolic IFS $\{\mathbb{R}^2; w_1, w_2\}$, where

$$\begin{aligned} w_1 \begin{pmatrix} x \\ y \end{pmatrix} &= \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \\ w_2 \begin{pmatrix} x \\ y \end{pmatrix} &= \begin{pmatrix} 0.3 & -0.2 \\ 0.1 & 0.4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}. \end{aligned}$$

Since (using $(z, w) = (\lambda_1, \lambda_2)$ to emphasize familiarity)

$$w_1(x, y, z, w) = (zx, wy + 1)$$

is continuous from basic calculus, Theorem 11.1 in Chapter III holds, and the attractor varies continuously with λ .

2 Mandelbrot Sets for Pairs of Transformations

Let $P \subset \mathbb{R}^2$ be a parameter space corresponding to a family of fractals. That is, we have a function $A : P \rightarrow \mathcal{H}(\mathbf{X})$, so that each point $\lambda \in P$ corresponds to a set $A(\lambda) \in \mathcal{H}(\mathbf{X})$. One way to make a map is to color the parameter space according to whether or not $A(\lambda)$ is connected.

Theorem 2.1 *Let $\{\mathbf{X}; w_1, w_2\}$ be a hyperbolic IFS with attractor A . Let w_1 and w_2 be one-to-one on A . If*

$$w_1(A) \cap w_2(A) = \emptyset,$$

then A is totally disconnected. If

$$w_1(A) \cap w_2(A) \neq \emptyset,$$

then A is connected.

Proof Suppose that $w_1(A) \cap w_2(A) = \emptyset$. Let Σ denote the code space map associated with the IFS. By Theorem 2.2 in Chapter IV the code space map $\phi : \Sigma \rightarrow A$ is invertible. ϕ is also a continuous transformation between two compact metric spaces.

Hence, by Theorem 8.5 in Chapter II, ϕ is a homeomorphism. Hence A is homeomorphic to code space, which is totally disconnected. (Recall that code space on two or more symbols is metrically equivalent to a classical Cantor set.) It follows that A is totally disconnected.

Suppose that $w_1(A) \cap w_2(A) \neq \emptyset$. Then there is at least one point $x \in w_1(A) \cap w_2(A)$. This point x has two addresses, say

$$x = \phi(\zeta) = \phi(\sigma), \quad \text{where } \zeta_1 = 1 \text{ and } \sigma_1 = 2.$$

Let us see what happens if we additionally suppose that “ A is not connected.” Then, since A is compact, we can find two nonempty compact sets E and F so that

$$A = E \cup F, \quad E \cap F = \emptyset.$$

Using compactness, there is a positive real number δ so that

$$d(e, f) \geq \delta \quad \text{for all } e \in E, f \in F.$$

Let π and ψ be a pair of codes that agree through the first K symbols, for some positive integer K . That is $\pi_i = \psi_i$ for $i = 1, 2, \dots, K$. Then

$$d(\phi(\pi), \phi(\psi)) \leq s^K \text{diam}(A),$$

where $\text{diam}(A) = \text{Max}\{d(x, y) : x, y \in A\}$, and $s \in [0, 1)$ is a contractivity factor for the IFS. Suppose also that $\phi(\pi) \in E$ and $\phi(\psi) \in F$. Then

$$\delta \leq d(\phi(\pi), \phi(\psi)).$$

Combining the latter two inequalities we discover $\delta \leq s^K \text{diam}(A)$, which implies

$$K \leq \frac{\log(\delta/\text{diam}(A))}{\log(s)}.$$

We conclude that if $e \in E$ and $f \in F$ then the number of successive agreements between an address of e and an address of f cannot exceed the number on the right-hand side here. It follows that there is a maximum number, M , of initial agreements between the address of a point $e \in E$ and a point $f \in F$; and this maximum is achieved on some pair of points, say e and f . Then we can find $\rho_i \in \{1, 2\}$ for $i = 1, 2, \dots, M$ such that

$$\phi(\rho_1, \rho_2, \rho_3, \dots, \rho_M, 1, \dots) = e \in E$$

and

$$\phi(\rho_1, \rho_2, \rho_3, \dots, \rho_M, 2, \dots) = f \in F.$$

Now consider the point $z \in A$, which has the two addresses

$$\begin{aligned} z &= \phi(\rho_1, \rho_2, \rho_3, \dots, \rho_M, 1, \zeta_2, \zeta_3, \zeta_4, \dots) \\ &= \phi(\rho_1, \rho_2, \rho_3, \dots, \rho_M, 2, \sigma_2, \sigma_3, \sigma_4, \dots). \end{aligned}$$

Suppose $z \in E$. Then its address agrees with that of $f \in F$ through $(M + 1)$ initial symbols. Hence $z \in F$. But then its address agrees with that of $e \in E$ through $(M +$

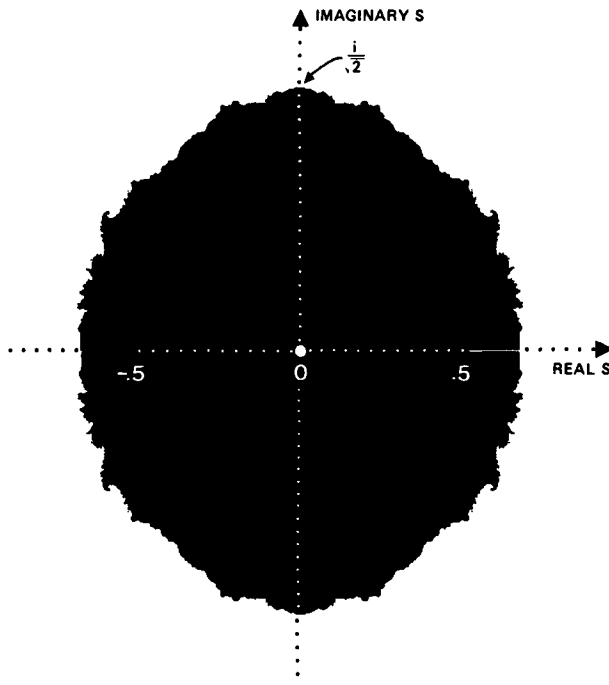


Figure VIII.221. A map of the family of IFS $\{\mathbb{C}; \lambda z - 1, \lambda z + 1\}$, where the parameter space is $P = \{\lambda = (\lambda_1, \lambda_2) \in \mathbb{C} : |\lambda| < 1\}$. This picture of a parameter space is obtained by “painting” black where the attractor of the IFS is disconnected and light where it is connected. The Mandelbrot set is the light region, the sea. It contains a dragon at $\lambda = (0.5, 0.5)$.

1) initial symbols, which is not possible. We have a contradiction. Hence “ A is not disconnected.” It follows that A is connected. This completes the proof of the theorem.

Definition 2.1 Let $\{X; w_1, w_2\}$ be a family of hyperbolic IFS that depends on a parameter $\lambda \in P \subset \mathbb{R}^2$. Let $A(\lambda)$ denote the attractor of the IFS. The set of points $\mathcal{M} \subset P$ defined by

$$\mathcal{M} = \{\lambda \in P : A(\lambda) \text{ is connected}\}$$

is called the Mandelbrot set for the family of IFS.

For the rest of this section we consider the family of IFS

$$\{\mathbb{C}; \lambda z - 1, \lambda z + 1\},$$

where the parameter space is

$$P = \{\lambda = (\lambda_1, \lambda_2) \in \mathbb{C} : \lambda_1^2 + \lambda_2^2 < 1\}.$$

Figure VIII.221 shows a picture of the associated Mandelbrot set, $\mathcal{M}$. This is a map for the collection of fractals associated with the IFS. It has been colored dark where the attractor is totally disconnected and light where it is connected.

Here is an outline of an algorithm to compute images of the Mandelbot set $\mathcal{M}$ associated with the family $\{\mathbb{C}; w_1(z) = \lambda z - 1, w_2(z) = \lambda z + 1\}$. It is based on Theorem 2.1.

Algorithm 2.1 Example of Method for Making Pictures of the Mandelbrot Set of a Family of IFS.

- (1) Choose a positive integer, L , corresponding to the amount of computation one is able to do. The greater the value of L , the more accurate the resulting map image will be.
- (2) Represent the parameter space $P = \{\lambda \in \mathbb{C} : |\lambda| < 1\}$ by an array of pixels. Carry out the following steps for each λ in the array.
- (3) Calculate a number R , so that the attractor is contained in a ball of radius R , centered at the origin; that is, choose $R > 0$ so that $A(\lambda) \subset B(0, R)$.
- (4) Compute the number

$$H = \min\{d(x, y) : x \in w_1(W^{\circ L}(\{0\})), y \in w_2(W^{\circ L}(\{0\}))\},$$

where $W = w_1 \cup w_2$. If $H \leq 2|\lambda|^{L+1}R$, then the pixel λ is assumed to belong to $\mathcal{M}$ and is colored accordingly.

Step (4) is based on the following observation. The attractor of the IFS is contained in the set $W^{\circ(L+1)}(B(0, R))$, which consists of 2^{L+1} balls of radius $|\lambda|^{L+1}R$. The centers of these balls lie in the union of the two sets $w_1(W^{\circ L}(0))$ and $w_2(W^{\circ L}(0))$. If H is greater than $2|\lambda|^{L+1}R$, then $A(\lambda)$ must be disconnected.

Figure VIII.222 shows the “coastal region” of a quarter of the complement of the Mandelbrot set in Figure VIII.221. It has been laid over a grid in order to help you locate points where interesting fractals lie.

The boundary of $\mathcal{M}$ is complicated and intricate. Close-ups of the “coastline” near the places marked (a), (b), and (c) are shown in Figure VIII.223. Figure VIII.224 shows a zoom on the spiral peninsula in Figure VIII.222.

Now look at Figures VIII.225 and VIII.226, which show pictures of the attractors $A(\lambda)$ for some points λ located near the boundary of the Mandelbrot set. There is a “family resemblance” between the places on the boundary from which the fractals come, and the fractals themselves. To help see this you should look back at the close-ups on the coastline in Figure VIII.223. Figure VIII.227 shows the IFS attractor corresponding to the tip of the peninsula in Figure 224. Notice how it contains spirals, very much like the ones in the peninsula in parameter space. At the end of this chapter we make some comments on why such “family resemblances” occur.

The following theorem provides rigorous bounds on the locations of $\mathcal{M}$ and $\partial\mathcal{M}$. The proof is delightful, because it relies on a fractal dimension estimate.

Theorem 2.2 [Barnsley 1985c] *The attractor $A(\lambda)$ of the IFS $\{\mathbb{C}; \lambda z - 1, \lambda z + 1\}$ is totally disconnected if $|\lambda| < 0.5$ and connected if $1 > |\lambda| > 1/\sqrt{2}$. The boundary of the associated Mandelbrot set is contained in the annulus $1/2 \leq |\lambda| \leq 1/\sqrt{2}$.*

Proof Let A denote the attractor of the IFS and let $D(A)$ denote its fractal dimension. The two maps in the IFS are similitudes of scaling factor $|\lambda|$. This means that Theorem 2.3 in Chapter V can be applied.

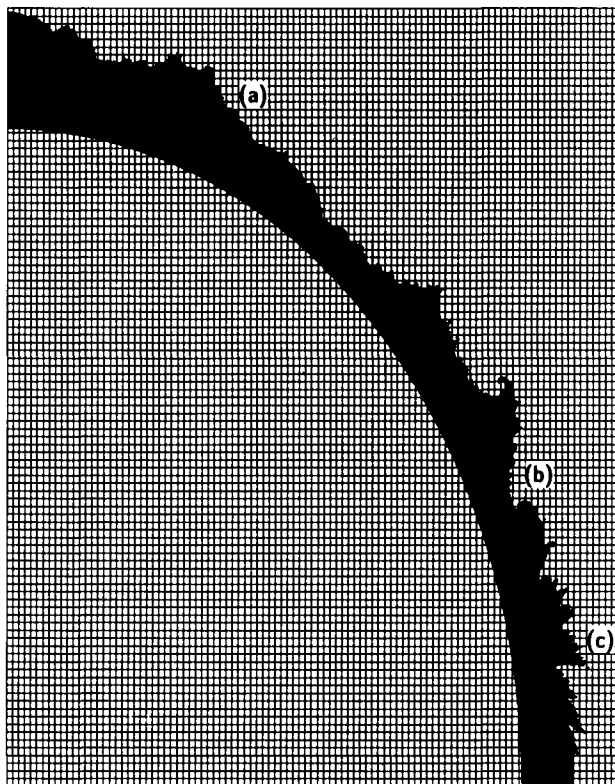


Figure VIII.222. This shows the coastal region of a quarter of the complement of the Mandelbrot set in Figure VIII.221. It has been laid over a grid in order to help you locate points where interesting fractals lie. Close-ups of the coast at (a), (b), and (c) are shown in Figure VIII.223. The coordinates of the grid are $(0, 0) - (0.71, 0.71)$.

Suppose that A is totally disconnected. Then the IFS is totally disconnected and, by Theorem 2.3 in Chapter V,

$$D(A) = \log(1/2) / \log(|\lambda|).$$

By Theorem 2.1 in Chapter V, $D(A) \leq 2$. Hence

$$\log(1/2) / \log(|\lambda|) \leq 2.$$

This implies that $|\lambda| \leq 1/\sqrt{2}$.

Suppose that A is connected. Then it contains a path that connects two distinct points. The fractal dimension of any path is greater than or equal to 1. Hence $D(A) \geq 1$. However, by Theorem 2.3 in Chapter V,

$$D(A) \leq \log(1/2) / \log(|\lambda|).$$

It follows that

$$1 \leq \log(1/2) / \log(|\lambda|).$$

This implies that $|\lambda| \geq 1/2$. This completes the proof of the theorem.

A different point of view on the Mandelbrot set considered above is given by

Figure VIII.223. Close-ups of the boundary of the Mandelbrot set at (a), (b), and (c). The diverse structures in this boundary echo the shapes of the attractors of the corresponding IFS.

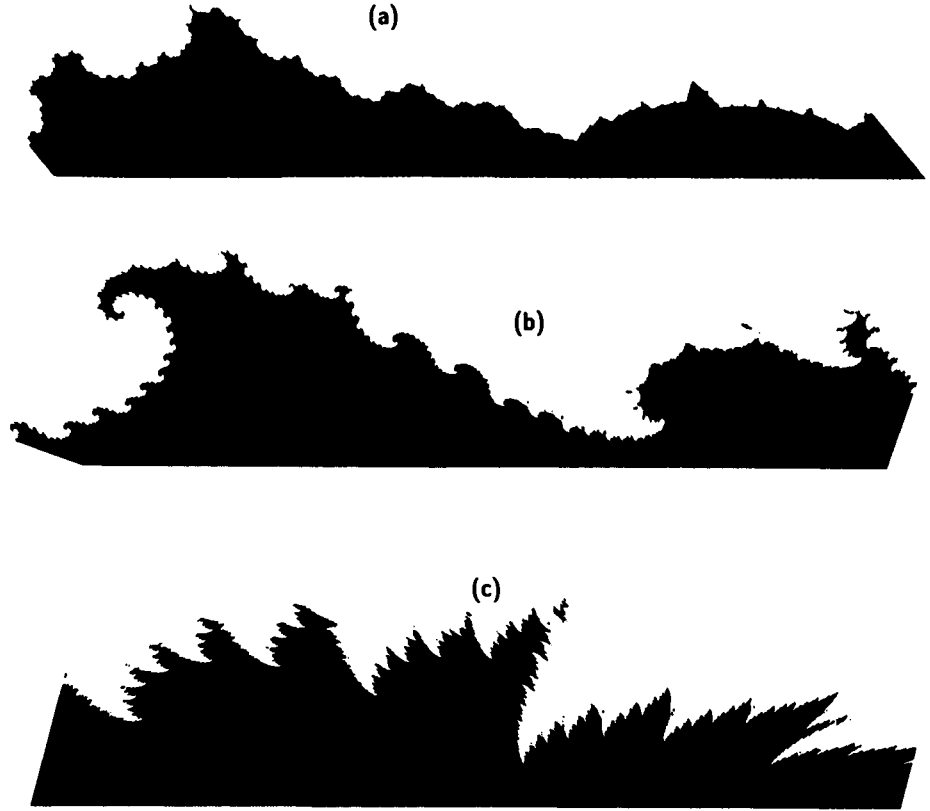


Figure VIII.228. $\mathcal{M}$ has been turned inside-out by making the change of variables $\lambda' = \lambda^{-1}$. The inner white disk is no-man's land; it does not belong to the parameter space. Also included are the two bounds provided by Theorem 2.2, namely the circle $|\lambda'| = 2$ and the circle $|\lambda'| = \sqrt{2}$. The fractal dimension decreases with increasing distance from the origin.

Examples & Exercises

2.1. Sketch the Mandelbrot set for the family of IFS $\{\mathbb{R}; \lambda_1 x + \lambda_2, \lambda_2 x + \lambda_1\}$ where the parameter space is $P = \{(\lambda_1, \lambda_2) : |\lambda_1|, |\lambda_2| < 1\}$.

2.2. Let $\{X; w_1, w_2\}$ be a family of hyperbolic IFS which depends on a parameter $\lambda \in P \subset \mathbb{R}^2$. Let w_1 and w_2 depend Lipschitz continuously on λ for some $k > 0$ and for fixed $x \in X$. Assume that the IFS has contractivity factor $s \in [0, 1)$ which is independent of $\lambda \in P$. Then by Theorem 11.1 in Chapter III, the function $A : P \rightarrow \mathcal{H}(X)$ is continuous. Use this continuity to prove that the Mandelbrot set associated with the family of IFS is closed. It is suggested that you begin by showing that the set $S = \{B \in \mathcal{H}(X) : B \text{ is not connected}\}$ is an open subset of $\mathcal{H}(X)$.

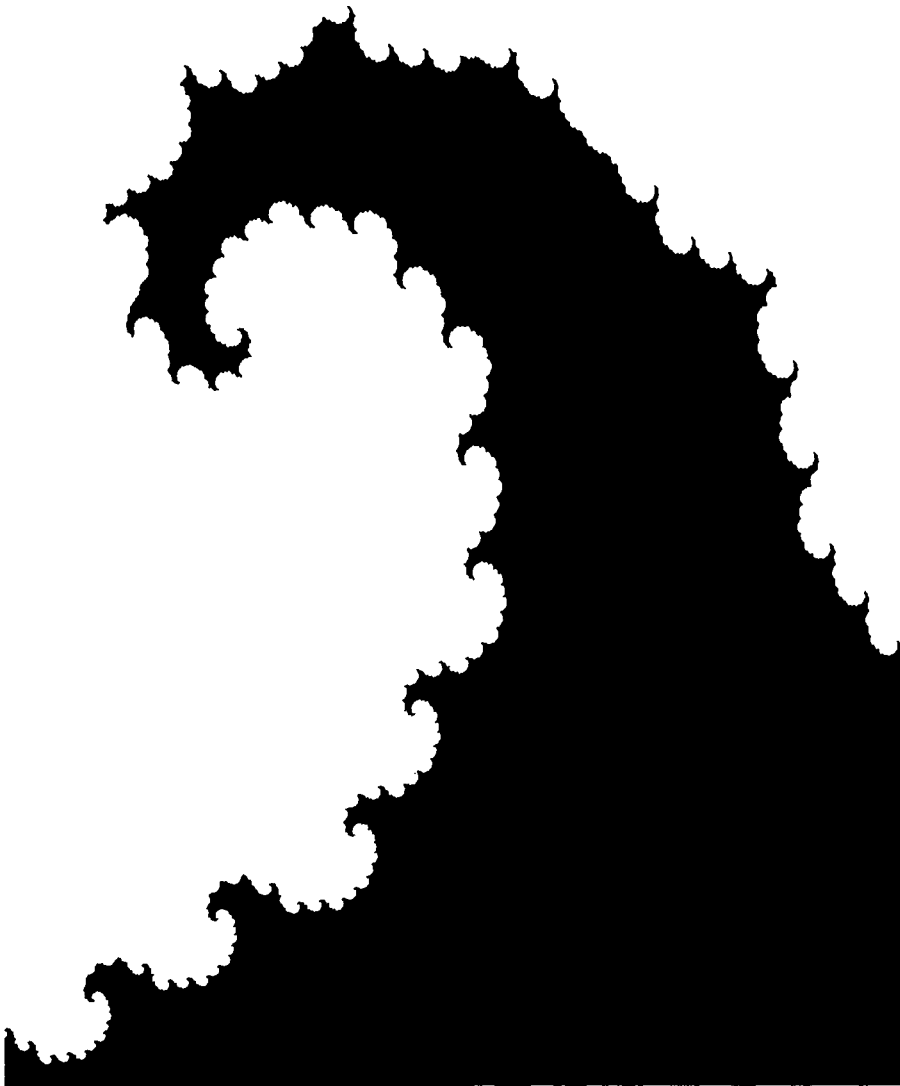
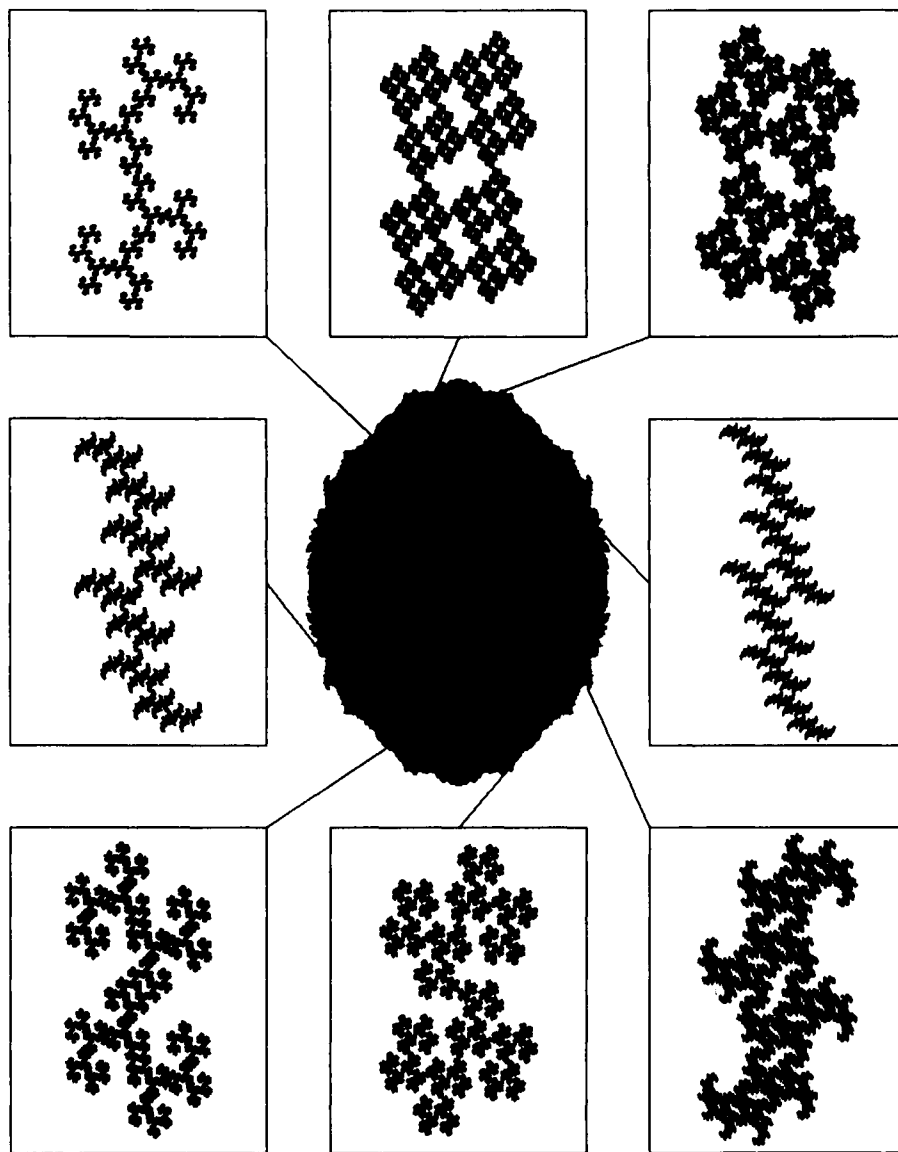


Figure VIII.224. Close-up on the spiral peninsula on the edge of the Mandelbrot set in Figures VIII.221 and VIII.222. What information about the corresponding fractals does this boundary convey?

2.3. Use Figure VIII.222 to determine some values of λ that belong, approximately, to the boundary $\partial\mathcal{M}$ of the Mandelbrot set. Compute images of the corresponding attractors. Compare images corresponding to two points λ_1 and $\lambda_2 \in \partial\mathcal{M}$, with $|\lambda_1| < |\lambda_2|$. Explain why the picture of $A(\lambda_1)$ is more delicate than the picture of $A(\lambda_2)$. Also comment on similarities and differences between your images and the local geography of the parts of $\partial\mathcal{M}$ from which they come.

2.4. The pictures of the Mandelbrot set associated with the family of IFS $\{\mathbb{C}; \lambda z -$

Figure VIII.225. Some of the fractals to be found at various points near the boundary of the Mandelbrot set associated with the parameterized family of IFS $\{\mathbb{C}; \lambda z - 1, \lambda z + 1\}$.



$1, \lambda z + 1\}$ suggest the conjecture that $\mathcal{M}$ is symmetric about the x -axis and about the origin. Prove the conjecture.

2.5. An interesting point in the parameter space for the family $\{\mathbb{C}; \lambda z - 1, \lambda z + 1\}$ is $\lambda = (1/2, 1/2)$. This lies on the circle $1/|\lambda| = |\lambda'| = \sqrt{2}$ in Figure VIII.228. It appears to be located in the interior of the Mandelbrot set, although the IFS is just-touching. It corresponds to the Twin-Dragon Fractal. A picture of it is shown in Figure VIII.229. It is possible to tile the plane with Twin-Dragons. Various other

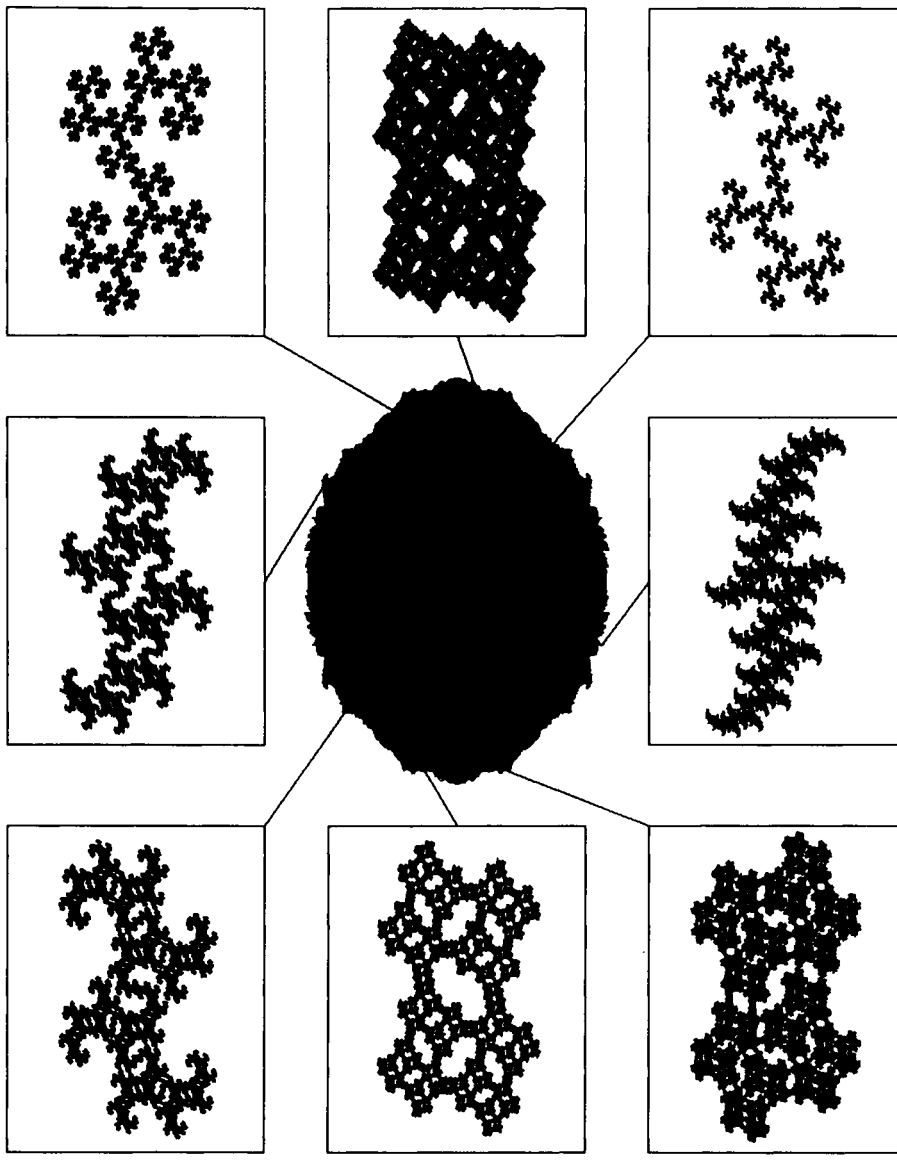


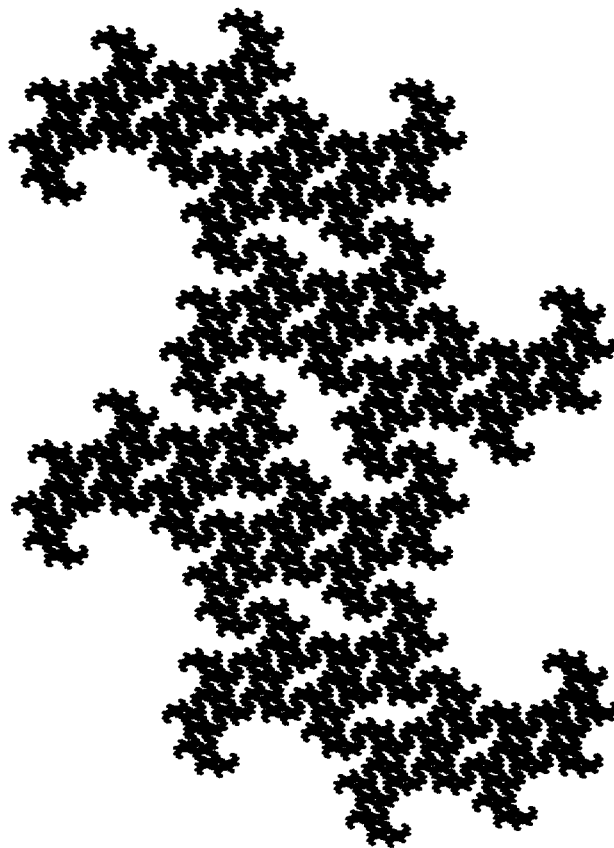
Figure VIII.226. Some of the IFS attractors to be found at various points near the boundary of the Mandelbrot set associated with the parameterized family of IFS $\{C; \lambda z - 1, \lambda z + 1\}$. Where would you look for an interesting fractal?

values of λ also correspond to tilings of the plane. See [Gilbert 1982]. Show that the attractor at the point $\lambda = (0, 1/\sqrt{2})$ can be used to tile the plane.

2.6. Notice the line segments on the real axis in Figure VIII.228. In [Barnsley 1985c] it is proved that

$$\{\lambda \in \mathbb{C} : 0.5 \leq \lambda_1 \leq 0.53; \lambda_2 = 0\} \subset \mathcal{M}.$$

Figure VIII.227. Attractor of the IFS $\{\mathbb{C}; \lambda z - 1, \lambda z + 1\}$ corresponding to the value of λ at the tip of the spiral peninsula, shown in Figure VIII.224.



For λ in such a line segment what does the attractor look like? Are you surprised, in view of what you know about maps of coastlines?

2.7. Some of the most delicate attractors of the family $\{\mathbb{C}; \lambda z - 1, \lambda z + 1\}$ are associated with points on $\partial\mathcal{M}$ where it touches the circle $1/|\lambda| = |\lambda'| = 2$. These have the lowest possible fractal dimension while still being connected. Let us call these attractors *tree-like* if $w_1(A) \cap w_2(A)$ is a single point. Argue (or, better yet, prove) that a tree-like attractor A contains no trapped holes; that is, A contains no nontrivial, non-self-intersecting paths that start and finish at the same point. A picture of a tree-like attractor is shown in Figure VIII.230.

2.8. Let $e = e_1 e_2 e_3 \dots e_n \dots$ be a point in the code space Σ of two symbols, with $e_n \in \{+1, -1\}$ for all n . Let $\lambda \in \mathbb{C}$. Prove that the series

$$f(\lambda) = e_1 + e_2 \lambda + e_3 \lambda^2 + e_4 \lambda^3 + e_5 \lambda^4 + e_6 \lambda^5 \dots + e_n \lambda^n + \dots$$

has radius of convergence 1. What is the relationship between $f(\lambda)$ and the code space map $\phi : \Sigma \rightarrow A(\lambda)$ associated with the family of IFS $\{\mathbb{C}; \lambda z - 1, \lambda z + 1\}$? Let

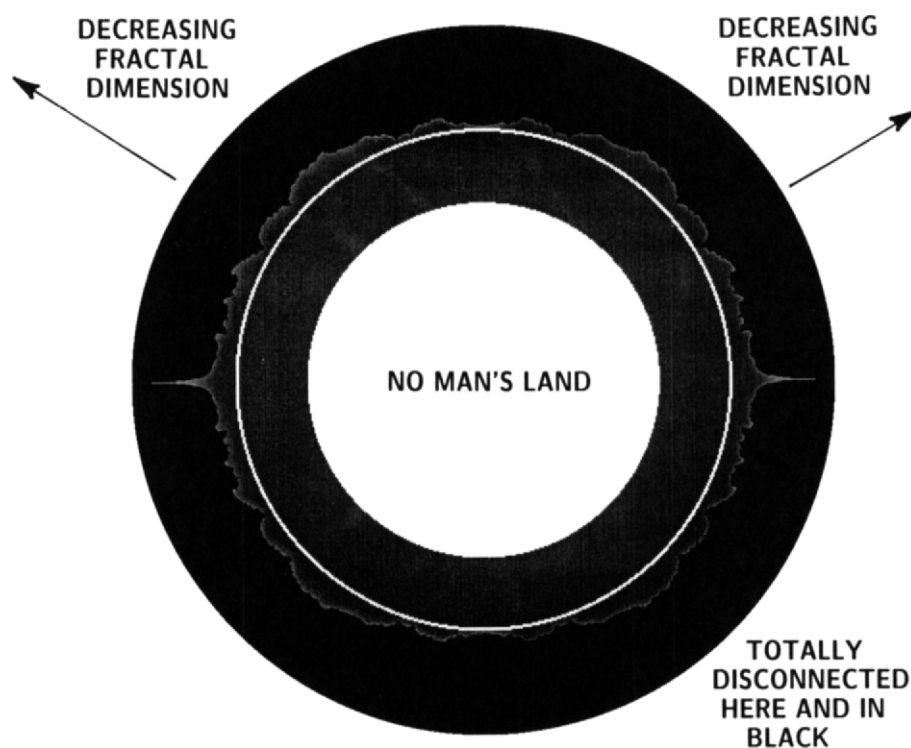


Figure VIII.228. Inside-out picture of the Mandelbrot set for $\{\mathbb{C}; \lambda z - 1, \lambda z + 1\}$. It has been turned inside-out by making the change of variables $\lambda' = \lambda^{-1}$. The inner white disk is no-man's land; it does not belong to the parameter space. The figure also includes the two bounds provided by Theorem 2.2, namely the circle $|\lambda'| = 2$ and the circle $|\lambda'| = \sqrt{2}$.

$|\lambda| < 1$. Show that the attractor of the IFS is the union of all points which can be written in the form

$$\pm 1 \pm \lambda \pm \lambda^2 \pm \lambda^3 \pm \lambda^4 \pm \lambda^5 \pm \lambda^6 \pm \lambda^7 \pm \lambda^8 \pm \lambda^9 \pm \lambda^{10} \pm \lambda^{11} \pm \dots$$

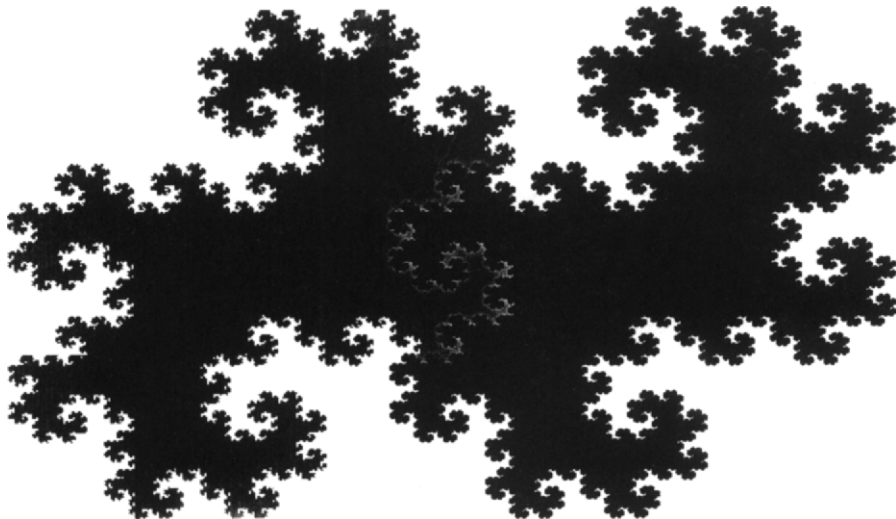
3 The Mandelbrot Set for Julia Sets

In this section we introduce a good method for making maps, such as might be found in an atlas, of families of dynamical systems. The method is based on the use of escape times and is discussed more generally in section 4. Here we restrict attention to the family

$$\{\hat{\mathbb{C}}; f_\lambda(z) = z^2 - \lambda\},$$

where the parameter space is $P = \mathbb{C}$. This family is of special importance because it provides a model for the onset of chaotic behavior in physical and biological systems; see [May 1976] and [Feigenbaum 1979]. Moreover, it was the first family of dynamical systems for which a useful computergraphical map was constructed, by Mandelbrot. We concentrate on map making.

Figure VIII.229. The Twin-Dragon Fractal. You can tile the plane with these sets. Although it is just-touching, it appears to lie in the interior of the Mandelbrot set.



The Julia set $J(\lambda)$ associated with $f_\lambda(z)$ is symmetric about the origin, O . We know this because the filled Julia set, of which $J(\lambda)$ is the boundary, is the set of points whose orbits remain bounded. The orbit of $z \in \mathbb{C}$ remains bounded if and only if the orbit of $-z$ remains bounded.

For some values of $\lambda \in P$, O belongs to the filled Julia set, $F(\lambda)$, while for others it is quite far from $F(\lambda)$. This suggests that we try to color the parameter space according to the distance from O to $F(\lambda)$. How can we estimate this distance? An approximate method is to look at the “escape time” of the orbit of O . That is, we can color the parameter space according to the number of steps along the orbit of O that are required before it lands in a ball around the point at infinity, from where we know that all orbits diverge. The intuitive idea is that the longer an orbit of O takes to reach the ball, the closer O must be to $F(\lambda)$. Of course, if an orbit does not diverge then we know that $O \in F(\lambda)$.

Suppose that we want to make a map corresponding to a region $\mathcal{W} \subset P$. Here we choose

$$\mathcal{W} = \{\lambda = (\lambda_1, \lambda_2) \in \mathbb{C} : |\lambda_1|, |\lambda_2| \leq 2\}.$$

Let $R > 0$ and define

$$\mathcal{V}(R) = \{z \in \mathbb{C} : |z| > R\} \cup \{\infty\}.$$

Suppose

$$R > 0.5 + 0.25 + |\lambda|.$$

Then it is readily proved that the orbit $\{f_\lambda^{on}(z)\}$ diverges if and only if it intersects $\mathcal{V}(R)$. So if we choose $R = 10$ we are sure that, for all $\lambda \in \mathcal{W}$, the orbit $\{f_\lambda^{on}(O)\}$

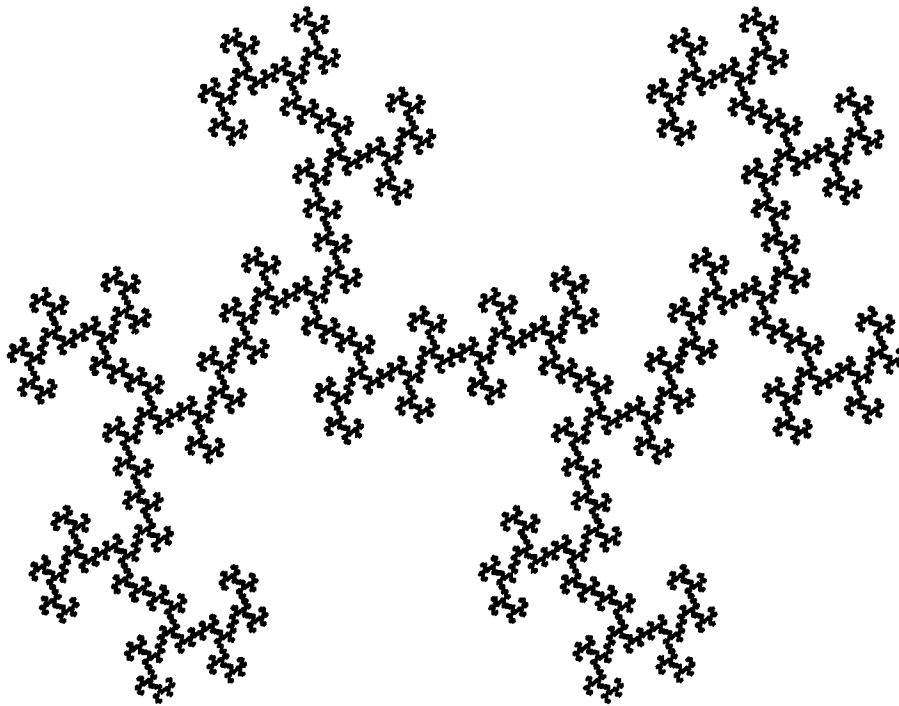


Figure VIII.230. A “tree-like” attractor A from the family $\{\mathbb{C}; w_1(z) = \lambda z - 1, w_2 = \lambda z + 1\}$. The two sets $w_1(A)$ and $w_2(A)$ meet approximately at a single point.

diverges if and only if it intersects $\mathcal{V}(R)$. Let us see what happens if we color the pixels of $\mathcal{W}$ according to the number of iterations required to enter $\mathcal{V}(10)$.

The following program is written in BASIC. It runs without modification on an IBM PC with Enhanced Graphics Adaptor and Turbobasic. On any line the words preceded by ' are comments: they are not part of the program.

Program 1. (Algorithm for Coloring Parameter Space According to an Escape Time)

```
'Define viewing window, W, and numits.
numits = 20 : a = -2 : b = -2 : c = 2 : d = 2 : M = 100

R =10 'Define the region V.

screen 9 : cls 'Initialize graphics.

for p =1 to M

for q =1 to M

'Specify the value of lambda (k,l) in P
k = a+(c-a)*p/M : l = b+(d-b)*q/M
```

```

'Specify the initial point, 0, on the orbit
x =0 : y =0

'Compute at most numits points on the orbit of 0
for n = 1 to numits

newx = x*x-y*y-k

newy = 2*x*y-l

x = newx : y = newy

'If the most recently computed point lies in V then...
if x*x+y*y > R then

'... render the pixel (p,q) in color n, and
'go to the next (p,q).
pset(p,q),n : n = numits

end if

if instat then end 'Stop computing if any key is pressed!

next n : next q : next p

end

```

Color Plate 16 shows the result of running a version of Program 1 on a Masscomp 5600 workstation with Aurora graphics.

In Figure VIII.231 we show the result of running a version of Program 1, but this time in halftones. The central white object corresponds to values of λ for which the computed orbit of O does not reach $\mathcal{V}$ during the first *numits* iterations. It represents the Mandelbrot set (defined below) for the dynamical system $\{\hat{C}; z^2 - \lambda\}$. The bands of colors (or white and shades of gray) surrounding the Mandelbrot set correspond to different numbers of iterations required before the orbit of O reaches $\mathcal{V}(10)$. The bands farthest away from the center represent orbits that reach O most rapidly. Approximately, the distance from O to $F(\lambda)$ increases with the distance from λ to the Mandelbrot set.

Definition 3.1 *The Mandelbrot set for the family of dynamical systems $\{\hat{C}; z^2 - \lambda\}$ is*

$$\mathcal{M} = \{\lambda \in P : J(\lambda) \text{ is connected}\}.$$

The relationship between escape times of orbits of O and the connectivity of $J(\lambda)$ is provided by the following theorem.

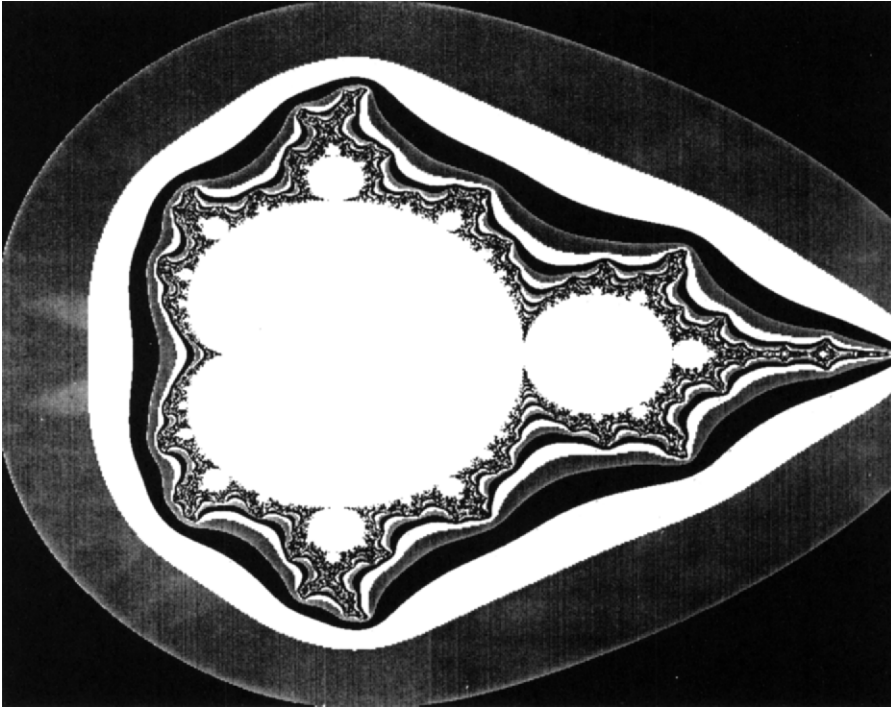


Figure VIII.231. The Mandelbrot set for $z^2 - \lambda$, computed by escape times.

Theorem 3.1 *The Julia set for the family of dynamical systems $\{\hat{\mathbb{C}}; f_\lambda(z) = z^2 - \lambda\}$, $\lambda \in P = \mathbb{C}$, is connected if and only if the orbit of the origin does not escape to infinity; that is*

$$\mathcal{M} = \{\lambda \in \mathbb{C} : |f_\lambda^{on}(0)| \not\rightarrow \infty \text{ as } n \rightarrow \infty\}$$

Proof This theorem follows from [Brolin], Theorem 11.2, which says that the Julia set of a polynomial, of degree greater than 1, is connected if and only if none of the finite critical points lie in the basin of attraction of the point at infinity. $f_\lambda(z)$ possesses two critical points, 0 and ∞ . Hence $J(\lambda)$ is connected if and only if $|f_\lambda^{on}(0)| \not\rightarrow \infty$ as $n \rightarrow \infty$.

In this paragraph we discuss the relationship between the Mandelbrot set for the family of dynamical systems $\{\hat{\mathbb{C}}; z^2 - \lambda\}$ and the corresponding family of IFS $\{\hat{\mathbb{C}}; \sqrt{z + \lambda}, -\sqrt{z + \lambda}\}$. We know that for various values of λ in $\mathbb{C}$ the IFS can be modified so that it is hyperbolic, with attractor $J(\lambda)$. For the purposes of this paragraph let us pretend that the IFS is hyperbolic, with attractor $J(\lambda)$, for all $\lambda \in \mathbb{C}$. Then Definition 2.1 would be equivalent to Definition 3.1. By Theorem 2.1, the attractor of the IFS would be connected if and only if $w_1(J(\lambda)) \cap w_2(J(\lambda)) \neq \emptyset$. But $w_1(\mathbb{C}) \cap w_2(\mathbb{C}) = 0$. Then it would follow that the attractor of the IFS is connected if and only if $0 \in J(\lambda)$. In other words: we discover the same criteria for connectivity

of $J(\lambda)$ if we argue informally using the IFS point of view, as can be proved using Julia set theory. This completes the discussion.

We return to the theme of coastlines and the possible resemblance between fractal sets corresponding to points on boundaries in parameter space and the local geometry of the boundaries. Figures VIII.232 and VIII.233 show the Mandelbrot set for $z^2 - \lambda$, together with pictures of filled Julia sets corresponding to various points around the boundary. If one makes a very high-resolution image of the boundary of the Mandelbrot set, at a value of λ corresponding to one of these Julia sets, one “usually” finds structures that resemble the Julia set. It is as though the boundary of the Mandelbrot set is made by stitching together microscopic copies of the Julia sets that it represents. An example of such a magnification of a piece of the boundary of $\mathcal{M}$, and a picture of a corresponding Julia set, are shown in Figures VIII.234 and VIII.235.

If you look closely at the pictures of the Mandelbrot set $\mathcal{M}$ considered in this section, you will see that there appear to be some parts of the set that are not connected to the main body. Pictures can be misleading.

Theorem 3.2 [*Mandelbrot-Douady-Hubbard*]

The Mandelbrot set for the family of dynamical systems $\{\hat{\mathbb{C}}; z^2 - \lambda\}$ is connected.

Proof This can be found in [Douady 1982].

The Mandelbrot set for $z^2 - \lambda$ is related to the exciting subject of cascades of bifurcations, quantitative universality, chaos, and the work of Feigenbaum. To learn more you could consult [Feigenbaum 1979], [Douady 1982], [Barnsley 1984], [Devaney 1986], [Peitgen 1986], and [Scia 1987].

Examples & Exercises

3.1. Rewrite Program 1 in a form suitable for your own computergraphical environment. Run your program and obtain hardcopy of the output. Adjust the window parameters a , b , c , and d , to allow you to make zooms on the boundary of the Mandelbrot set.

3.2. Figure VIII.236 shows a picture of the Mandelbrot set for the family of dynamical systems $\{\hat{\mathbb{C}}; z^2 - \lambda\}$ corresponding to the coordinates $-0.5 \leq \lambda_1 \leq 1.5$, $-1.0 \leq \lambda_2 \leq 1.0$. It has been overlaid on a coordinate grid. The middle of the first bubble has not been plotted, to clarify the coordinate grid. Let $B_0, B_1, B_2, B_3, \dots$ denote the sequence of bubbles on the real axis, reading from left to right. Verify computationally that when λ lies in the interior of B_n the dynamical system possesses an attractive cycle, located in $\mathbb{C}$, of minimal period 2^n , for $n = 0, 1, 2$, and 3.

3.3. The sequence of bubbles $\{B_n\}_{n=0}^{\infty}$ in exercise 3.2 converges to the *Myreberg point*, $\lambda = 1.40115\dots$. The ratios of the widths of successive bubbles converge to the *Feigenbaum ratio* $4.66920\dots$. Make a conjecture about what sort of “attractive cycle” the dynamical system $\{\hat{\mathbb{C}}; z^2 - \lambda\}$ might possess at the Myreberg point. Test your conjecture numerically. You will find it easiest to restrict attention to real orbits.

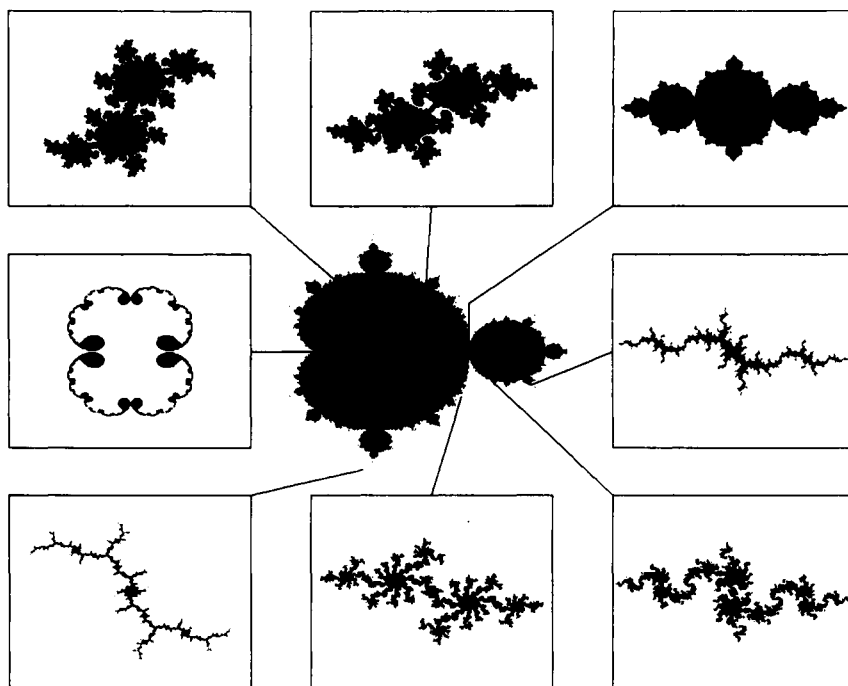


Figure VIII.232. Mandelbrot set for $z^2 - \lambda$, decorated with various Julia sets and filled Julia sets.

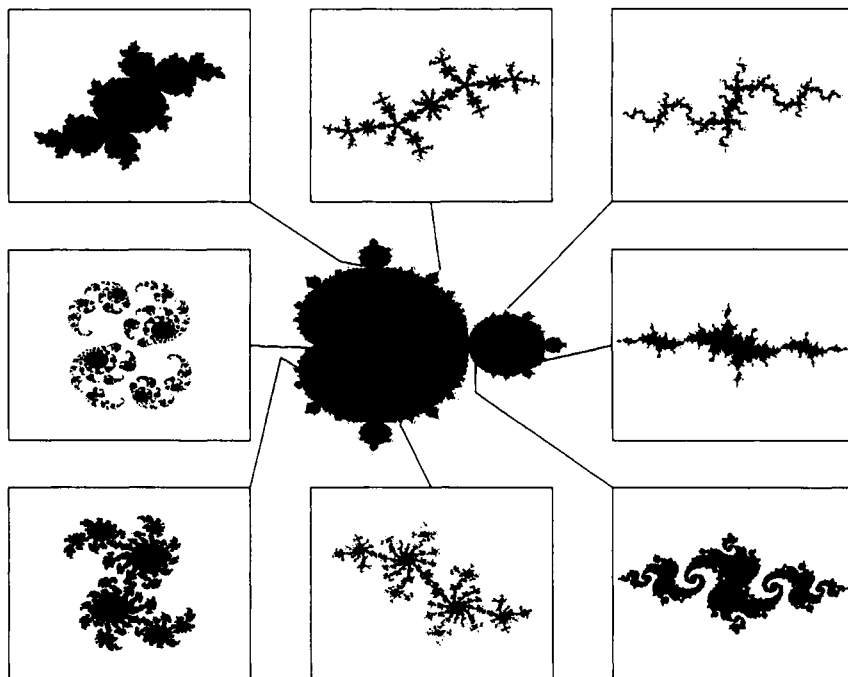


Figure VIII.233. Mandelbrot set for $z^2 - \lambda$, decorated with various Julia sets and filled Julia sets. These often resemble the place on the boundary from which they come, especially if one magnifies up enough.

Figure VIII.234. A zoom on a piece of the boundary of the Mandelbrot set for $z^2 - \lambda$.

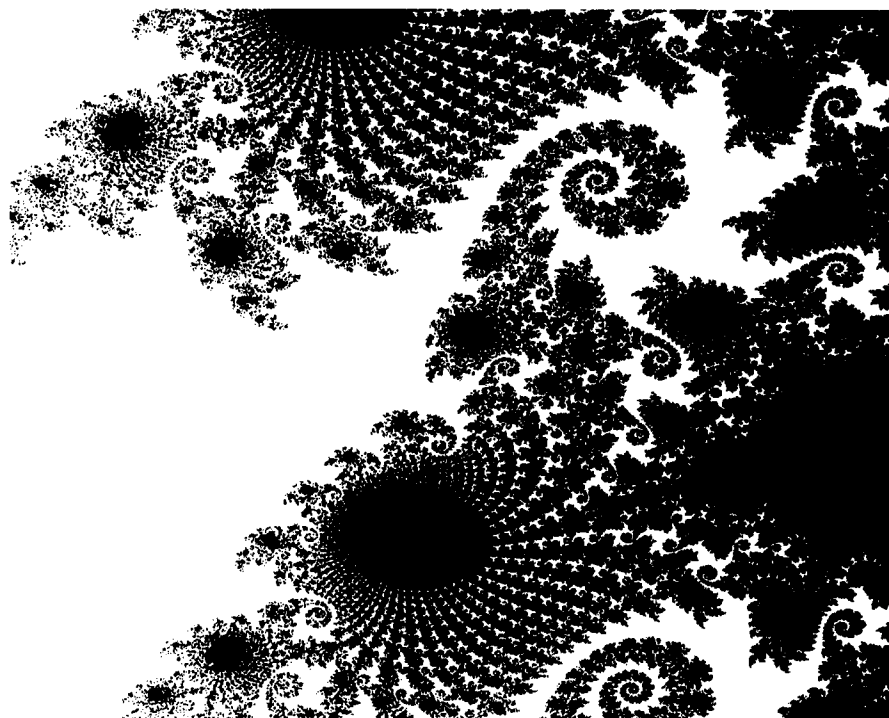
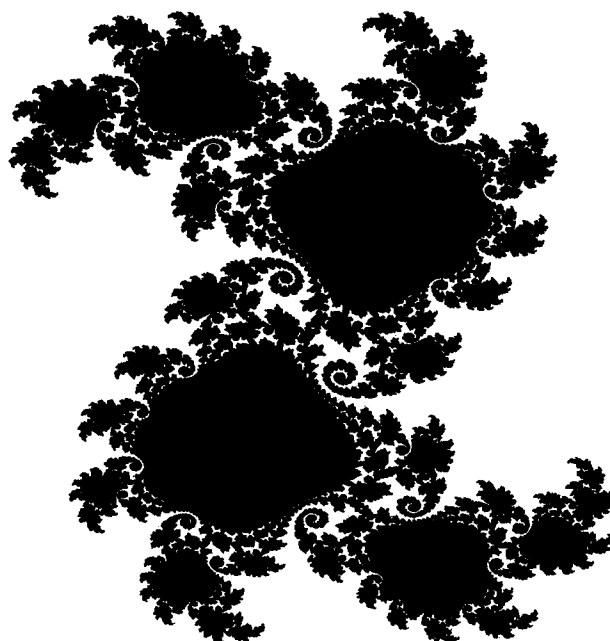


Figure VIII.235. A filled Julia set corresponding to the piece of the coastline of the Mandelbrot set in Figure VIII.234. Notice the family resemblances.



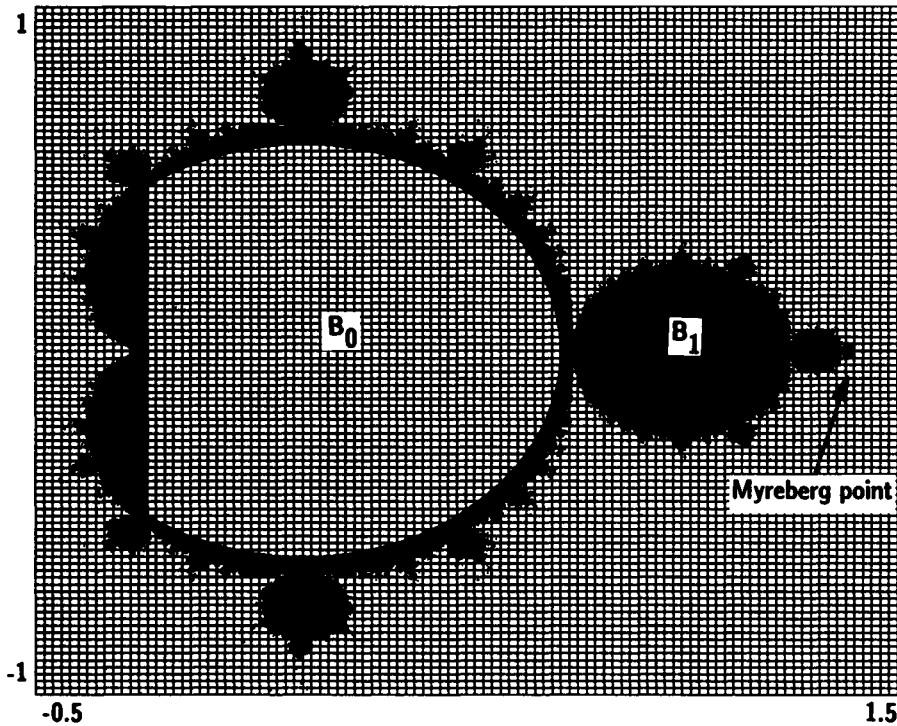


Figure VIII.236. A picture of the Mandelbrot set for the family of dynamical systems $\{\mathbb{C}; z^2 - \lambda\}$. It has been overlaid on a coordinate grid. The middle of the first bubble has not been plotted, to clarify the coordinate grid.

3.4. Make a parameter space map for the family of dynamical systems $\{\hat{\mathbb{C}}; f_\lambda(z)\}$, where f_λ is the Newton transformation associated with the family of polynomials

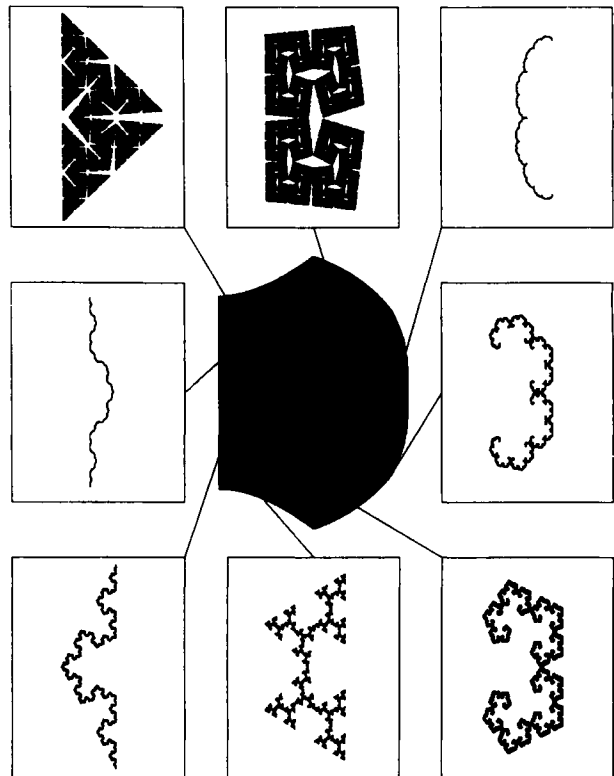
$$F(z) = z^3 + (\lambda - 1)z + 1, \quad \lambda \in P = \mathbb{C}.$$

Notice that the polynomial has a root located at $z = 1$, independent of λ . Color your map according to the “escape time” of the orbit of O to a ball of small radius centered at $z = 1$. Use black to represent values of λ for which O does not converge to $z = 1$. Examine some Julia sets of f_λ corresponding to points on the boundary of the black region. Are there resemblances between structures that occur in your map of parameter space, and some of the corresponding collection of Julia sets? (The correct answer to this question can be found in [Curry 1983].)

4 How to Make Maps of Families of Fractals Using Escape Times

We begin by looking at the Mandelbrot set for a certain family of IFS. It is disappointing, and we do not learn much. We then introduce a related family of dynamical systems and color the parameter space using escape times. The result is a map

Figure VIII.237. The complement of the Mandelbrot set $\mathcal{M}_1$ associated with the family of IFS $\{\mathbb{C}; w_1 = \lambda z + 1, w_2 = \lambda^* z - 1\}$. Points in the complement of the Mandelbrot set are colored black. The boundary of $\mathcal{M}_1$ is smooth and does not reveal much information about the family of fractals it represents. The figure also shows attractors of the IFS corresponding to various points on the boundary of $\mathcal{M}_1$. What a disappointing map this is!



packed with information. We generalize the procedure to provide a method for making maps of other families of dynamical systems. We discover how certain boundaries in the resulting maps can yield information about the appearance of the fractals in the family. That is, we begin to learn to read the maps.

Figures VIII.237 and VIII.238 show the Mandelbrot set $\mathcal{M}_1$ for the family of hyperbolic IFS

$$\{\mathbb{C}; w_1(z) = \lambda z + 1, w_2(z) = \lambda^* z - 1\}, P = \{\lambda \in \mathbb{C} : |\lambda| < 1\}.$$

We use the notation $\lambda^* = (\lambda_1 + i\lambda_2)^* = (\lambda_1 - i\lambda_2)$ for the complex conjugate of λ . The two transformations are similitudes of scaling factor $|\lambda|$. At fixed λ , they rotate in opposite directions through the same angle. The figures also show attractors of the IFS corresponding to various points around the boundary of the Mandelbrot set. What a disappointing map this is! There are no secret bays, jutting peninsulas, nor ragged rocks in the coastline.

Theorem 4.1 (Hardin 1985). *The Mandelbrot set $\mathcal{M}_1$ is connected. Its boundary is the union of a countable set of smooth curves and is piecewise differentiable.*

Proof This can be found in [Barnsley 1988d].

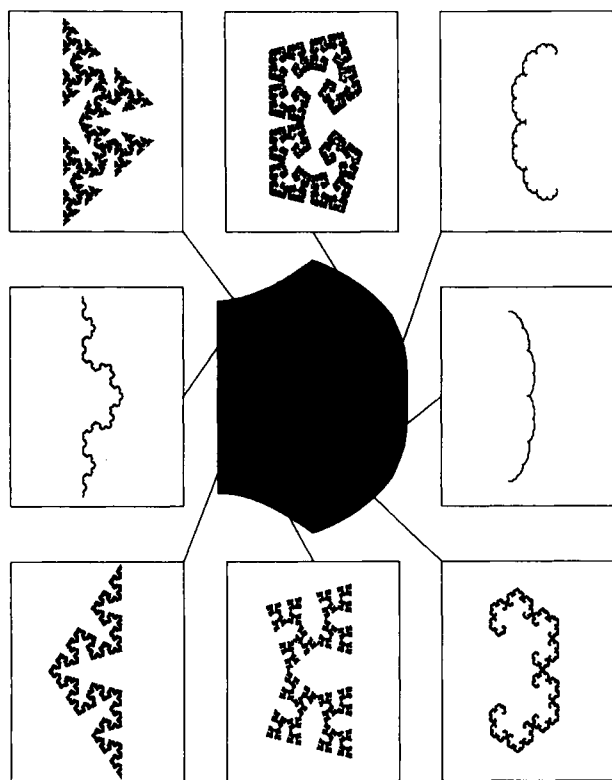


Figure VIII.238. The complement of the Mandelbrot set $\mathcal{M}_1$ associated with the family of IFS $\{\mathbb{C}; w_1 = \lambda z + 1, w_2 = \lambda^* z - 1\}$, together with some of the corresponding fractals. Notice how these have subsets of points that lie on straight lines, like the local structure of $\partial\mathcal{M}'_1$.

Let us try to obtain a better map of this family of attractors. In order to do so we begin by defining an extension of the associated shift dynamical system, for each $\lambda \in P \setminus \mathcal{M}_1$. Let $A(\lambda)$ denote the attractor of the IFS. One can prove that $A(\lambda)$ is symmetric about the y -axis. Hence $A(\lambda) \in \mathcal{M}_1$ if and only if $A(\lambda)$ intersects the y -axis. Define $f_\lambda : \mathbb{C} \rightarrow \mathbb{C}$

$$f_\lambda(z) = \begin{cases} w_1^{-1}(z) & \text{if } \operatorname{Re} z \geq 0; \\ w_2^{-1}(z) & \text{if } \operatorname{Re} z < 0. \end{cases}$$

Then, when λ is such that $A(\lambda)$ is disconnected, $\{A(\lambda); f_\lambda\}$ is the shift dynamical system associated with the IFS; $\{\mathbb{C}; f_\lambda\}$ is an extension of the shift dynamical system to all of $\mathbb{C}$; and $A(\lambda)$ is the “repelling set” of $\{\mathbb{C}; f_\lambda\}$. This system can be used to compute images of $A(\lambda)$ in the just-touching and totally disconnected cases, using the Escape Time Algorithm, as discussed in Chapter VII, section 1.

We make a map of the family of dynamical systems $\{\mathbb{R}^2; f_\lambda\}$, $\lambda \in P$. To do this we use the following algorithm, which was illustrated in Program 8.3.1. The algorithm applies to any family of dynamical systems $\{\mathbb{R}^2; f_\lambda\}$ that possesses a “repelling set” $A(\lambda)$, and such that P is a two-dimensional parameter space with a nice classical shape, such as a square or a disk.

Algorithm 4.1 Method for Coloring Parameter Space According to an Escape Time.

- (i) Choose a positive integer, *numits*, corresponding to the amount of computation one is able to do. Fix a point $Q \in \mathbb{R}^2$ such that $Q \in A(\lambda)$ for some, but not all, $\lambda \in P$.
- (ii) Fix a ball $B \subset \mathbb{R}^2$ such that $A(\lambda) \subset B$ for all $\lambda \in P$. Define an escape region to be $\mathcal{V} = \mathbb{R}^2 \setminus B$.
- (iii) Represent the parameter space P by an array of pixels. Carry out the following step for each λ in the array.
- (iv) Compute $\{f_\lambda^{on}(Q) : n = 0, 1, 2, 3, \dots, \text{numits}\}$. Color the pixel λ according to the least value of n such that $f_\lambda^{on}(Q) \in \mathcal{V}$. If the computed piece of the orbit does not intersect $\mathcal{V}$, color the pixel black.

The result of applying this algorithm to the dynamical system defined above, with $Q = O$, is illustrated in Figures VIII.239 and VIII.240 (a)–(g) and Color Plates 17 and 18.

Figure VIII.239. A map of the family of dynamical systems $\{C; f_\lambda\}$, where

$$f_\lambda(z) = \begin{cases} (z-1)/\lambda & \text{if } \operatorname{Re} z \geq 0; \\ (z+1)/\lambda & \text{if } \operatorname{Re} z < 0. \end{cases}$$

The parameter space is $P = \{\lambda \in \mathbb{C}; 0 < \lambda_1 < 1, 0 < \lambda_2 \leq 0.75\}$. The map is obtained by applying Algorithm 8.4.1. Pixels are shaded according to the “escape time” of a point $O \in \mathbb{R}^2$. The exciting places where the interesting fractals are to be found are not within the solid bands of black, gray, or white, but within the foggy coastline. This coastline is itself a fractal object, revealing infinite complexity under magnification. In it one finds approximate pictures of some of the connected and “almost connected” repelling sets of the dynamical system. Why are they there?

Figure VIII.239 contains four different regions. The first is a neighborhood of O , surrounded by almost concentric bands of black, gray, and white. The location of this region is roughly the same as that of $P \setminus \mathcal{M}_1$, which corresponds to totally disconnected and just-touching attractors. The second region is the grainy area, which

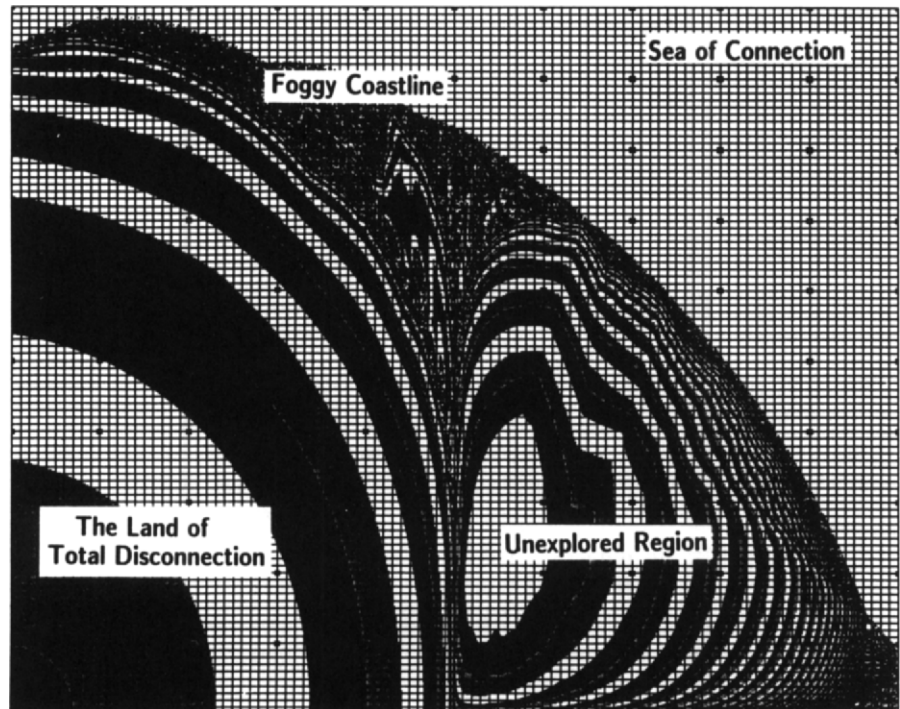




Figure VIII.240. A sequence of zooms on a piece of the foggy coastline in Figure VIII.239. The window coordinates of the highest power zoom are $0.4123 \leq \lambda_1 \leq 0.4139$, $0.6208 \leq \lambda_2 \leq 0.6223$. Can you find where each picture lies within the one that precedes it?

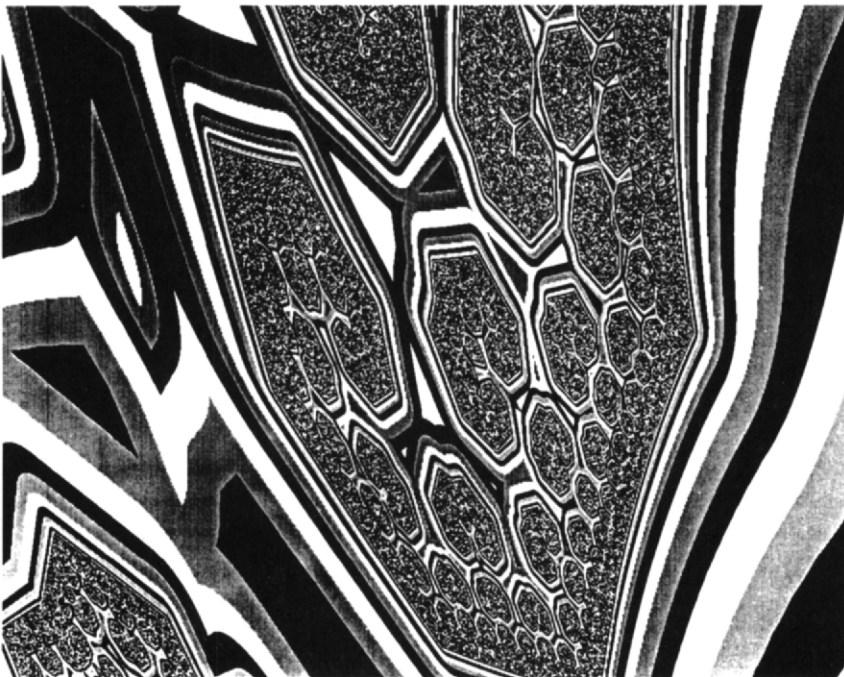


Figure VIII.240. (b)

Figure VIII.240. (c)

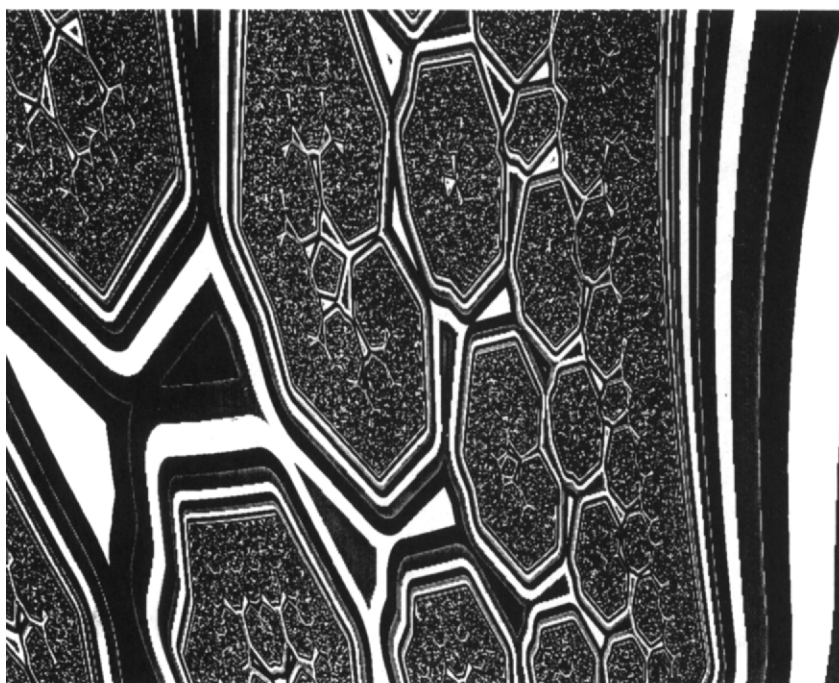
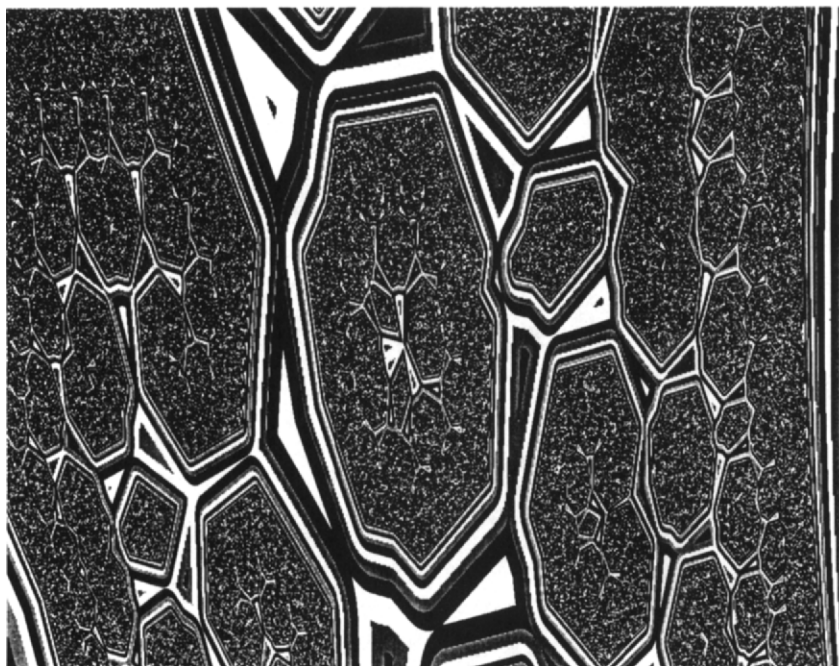


Figure VIII.240. (d)



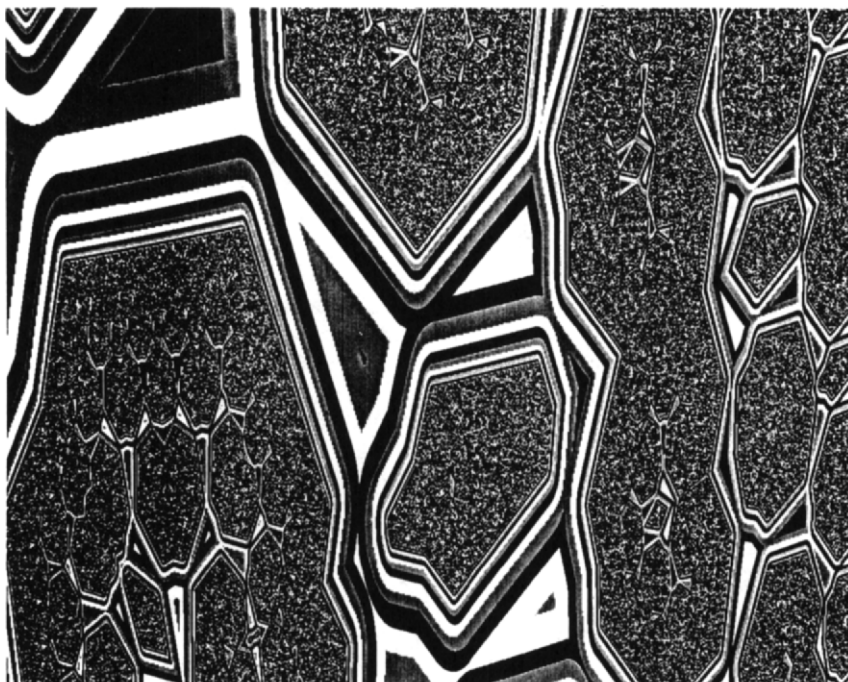


Figure VIII.240. (e)

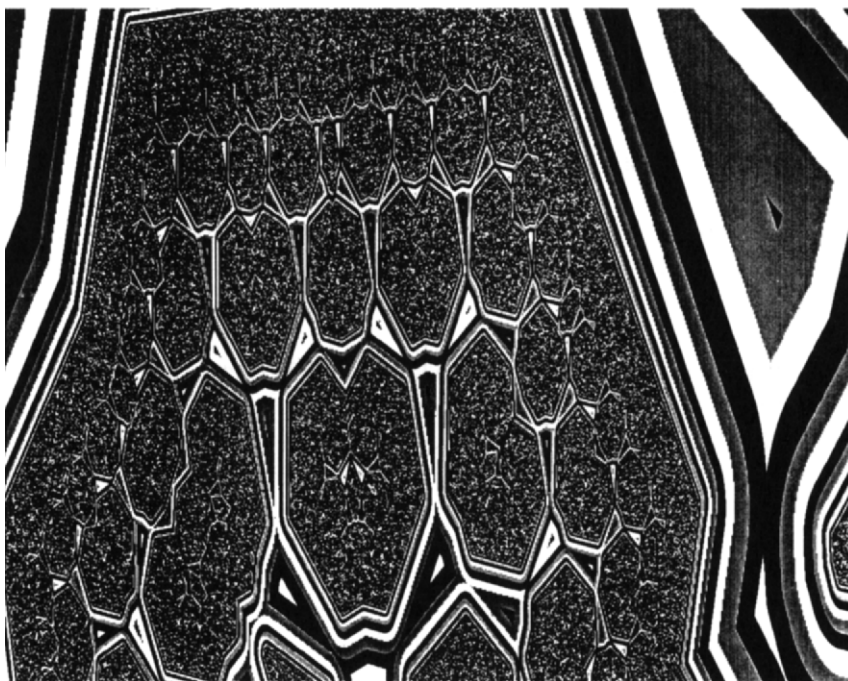
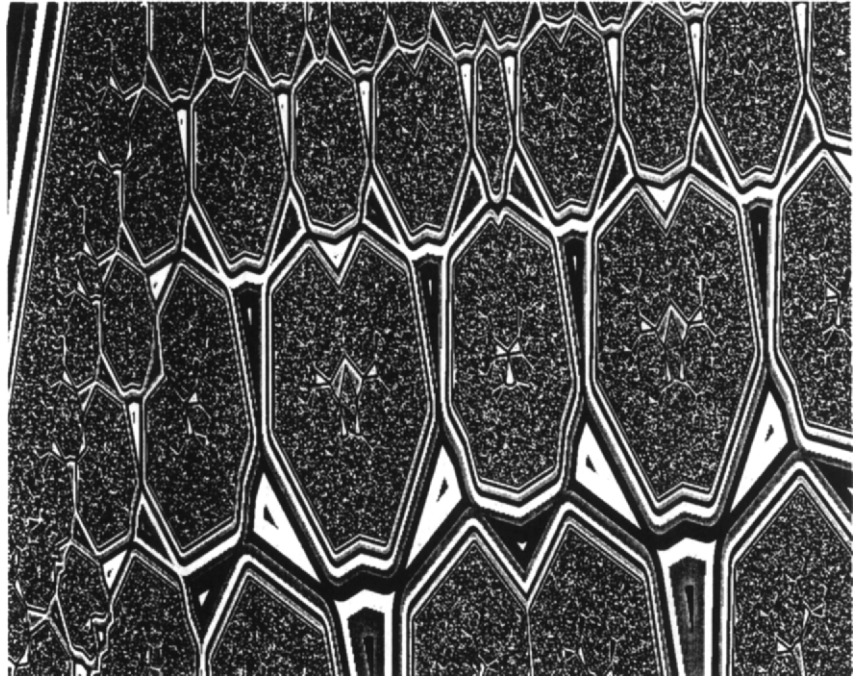


Figure VIII.240. (f)

Figure VIII.240. (g)



we refer to as the foggy coastline. Here, upon magnification, one finds complex geometrical structures. An example is illustrated in the sequence of zooms in Figure VIII.240 (a)–(g). The structures appear to be subtly different from one another. Early experiments show that if λ is chosen in the vicinity of one of these structures, then images of the “repelling set” of the dynamical system $\{\mathbb{R}^2; f_\lambda\}$, computed using the Escape Time Algorithm, contains similar structures. An example of such an image is shown in Figure VIII.241. The third region, at the lower right in Figure VIII.239, is made up of closed contours of black, grey, and white. Here the map conveys little information about the family of dynamical systems. To obtain information in this region one should examine the orbits of a point Q , different from O . The fourth region, the outer white area in Figure VIII.239, corresponds to dynamical systems for which the orbit of O does not escape. It is likely that for λ in this region, the “repelling set” of the dynamical system possesses an interior.

Our new maps, such as Figure VIII.239, can provide information about the family of IFS

$$\{\mathbb{C}; w_1(z) = \lambda z + 1, w_2(z) = \lambda^* z - 1\}, P = \{\lambda \in \mathbb{C} : |\lambda| < 1\}$$

in the vicinity of the boundary of the Mandelbrot set. For $\lambda \in \partial\mathcal{M}_1$ the attractor of the IFS is the same as the repelling set of the dynamical system. For λ close to $\partial\mathcal{M}_1$ the attractor of the IFS “looks like” the repelling set of the dynamical system.

Figure VIII.242 shows a transverse section through the anther of a lily. We include it because some of the structures in Figure VIII.240 (a)–(g) are reminiscent of cells.

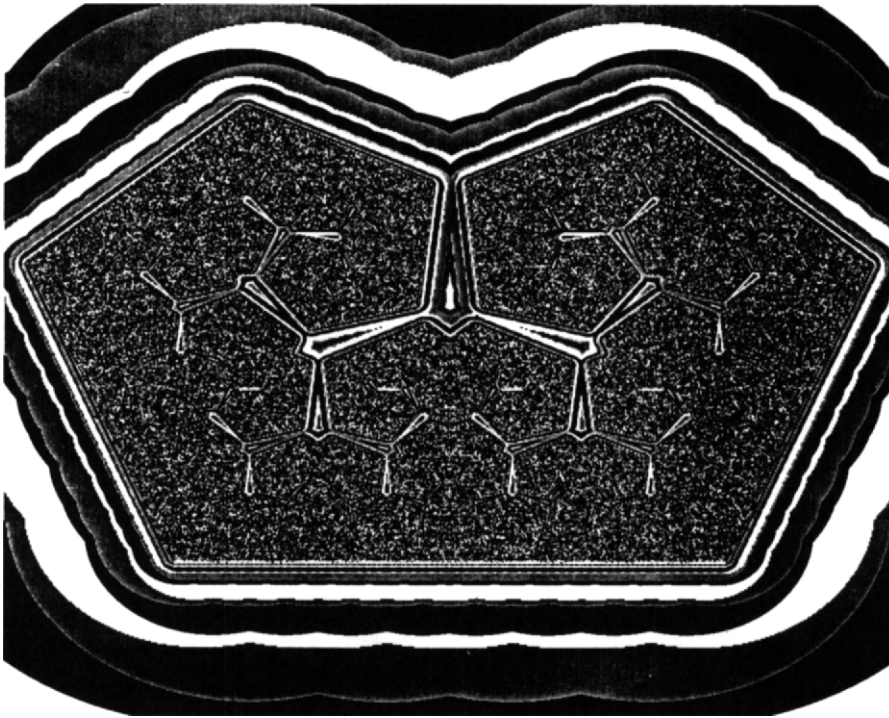


Figure VIII.241. Image of the repelling set for one of the family of the dynamical systems whose parameter space was mapped in Figure VIII.239. This image corresponds to a value of λ that lies within the highest power zoom in Figure VIII.240. Notice how the objects here resemble those in the corresponding position in the parameter space.

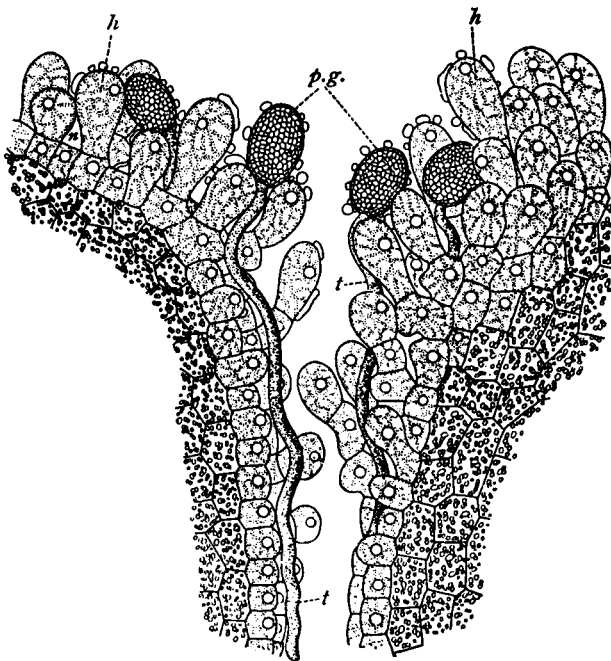


Figure VIII.242. Longitudinal section through part of the stigma of a lily, showing germinating pollen-grains. *h* papillae of stigma; *p.g.*, pollen grains; *t*, pollen tubes. Highly magnified. (After DodelPort, [Scott 1917].)

Algorithm 1 in section 4 can be applied to families of dynamical systems of the type described in Theorem 4.1 in Chapter VII. For example, let $\{\mathbb{R}^2; f_\lambda\}$, where $\lambda \in P = \mathbb{R}^2$, denote a family of dynamical systems. Let $X \subset \mathbb{R}^2$ be compact. Let $f_\lambda : X \rightarrow \mathbb{R}^2$ be continuous and such that $f(X) \supset X$. Then f_λ possesses an invariant set $A(\lambda) \in \mathcal{H}(X)$, given by

$$A(\lambda) = \bigcap_{n=0}^{\infty} f_\lambda^{\circ(-n)}(X).$$

$A(\lambda)$ is the set of points whose orbits do not escape from X . The set of points in P corresponding to which orbit of Q does not escape from X is

$$\mathcal{M}(Q) = \{\lambda \in P : Q \in A(\lambda)\}.$$

We conclude this chapter by giving an “explanation” of how family resemblances can happen between structures that occur on the boundary of $\mathcal{M}(Q)$ and the sets $A(\lambda)$. (1) Suppose that $A(\lambda)$ is a set in $\mathbb{R}^2$ that looks like a map of Great Britain, translated by λ . Then what does $\mathcal{M}(Q)$ look like? It looks like a map of Great Britain. (2) Suppose that $A(\lambda)$ is a set that looks like a map of Great Britain at time λ_1 , translated by λ . We picture the set $A(\lambda)$ varying slowly, perhaps its boundary changing continuously in the Hausdorff metric as λ varies. Now $A(\lambda)$ looks like a deformed map of Great Britain. The local coves and inlets will be accurate representations of those coves at about the time λ_1 to which they correspond in the parameter space map. That is, the boundary of $\mathcal{M}(Q)$ will consist of neighboring bays and inlets at different times stitched together. It will be a map that is microscopically accurate (at some time) and globally inaccurate. (3) Now pretend in addition that the coastline of Great Britain is self-similar at each time λ_1 . That is, imagine that little bays look like whole chunks of the coastline, at a given instant. Now what will $\mathcal{M}(Q)$ look like? At a given microscopic location on the boundary, magnified enormously, we will see a picture of a whole chunk of the coastline of Great Britain, at that instant. (4) Now imagine that for some values of λ , Great Britain, in the distant future, is totally disconnected, reduced to grains of isolated sand. It is unlikely that those values of λ belong to $\mathcal{M}(Q)$. As λ varies in a region of parameter space for which $A(\lambda)$ is totally disconnected, it is not probable that $Q \in A(\lambda)$. In these regions we would expect $\mathcal{M}(Q)$ to be totally disconnected.

The families of sets $\{A(\lambda) \in X : \lambda \in P\}$ considered in this chapter broadly fit into the description in the preceding paragraph. Both P and X are two-dimensional. The sets $A(\lambda)$ are derived from transformations that behave locally like similitudes. For each $\lambda \in P$, $A(\lambda)$ is either connected or totally disconnected. Finally, the sets $A(\lambda)$ and their boundaries appear to depend continuously on λ .

Examples & Exercises

4.1. In the above section we applied Algorithm 1 in section 4, with $Q = (0, 0)$, to compute a map of the family of dynamical systems

$$f_\lambda(z) = \begin{cases} (z-1)/\lambda & \text{if } \operatorname{Re} z \geq 0, \\ (z+1)/\lambda^* & \text{if } \operatorname{Re} z < 0. \end{cases}$$

The resulting map was shown in Figure VIII.239. This map contains an unexplored region. Repeat the computation, but with (a) $Q = 0.5$, and (b) $Q = -0.5$, to obtain information about the unexplored region.

4.2. In this example we consider the family of dynamical systems $\{\mathbb{C}; f_\lambda\}$, where

$$f_\lambda(z) = \begin{cases} (z-1)/\lambda & \text{if } \lambda_2 x - \lambda_1 y \geq 0, \\ (z+1)/\lambda & \text{if } \lambda_2 x - \lambda_2 y < 0. \end{cases}$$

The parameter space is $\lambda \in P = \{\lambda \in \mathbb{C} : 0 < |\lambda| < 1\}$. This family is related to the family of IFS

$$\{\mathbb{C}; w_1(z) = \lambda z + 1, w_2(z) = \lambda z - 1\}.$$

Let $A(\lambda)$ denote the attractor of the IFS and let $\tilde{A}(\lambda)$ denote the “repelling set” associated with the dynamical system. Let

$$S = \{\lambda \in P : \text{the line } \lambda_2 x - \lambda_1 y = 0 \text{ separates } w_1(A(\lambda)) \text{ and } w_2(A(\lambda))\}.$$

If $\lambda \in S$ then $\{A(\lambda); f_\lambda\}$ is the shift dynamical system associated with the IFS, and $A(\lambda) = \tilde{A}(\lambda)$. Even when $\lambda \notin S$ we expect there to be similarities between $A(\lambda)$ and $\tilde{A}(\lambda)$.

In Figures VIII.243, VIII.244, and VIII.245 and Color Plates 19 and 20 we show some results of applying Algorithm 1 in section 4 to the dynamical system $\{\mathbb{C}; f_\lambda\}$.

In Figure VIII.243, the outer white region represents systems for which the orbit of the point O does not diverge, and probably corresponds to “repelling sets” with nonempty interiors. The inner region, defined by the patchwork of gray, black, and white sections, bounded by line segments, represents systems for which the orbit of O diverges and corresponds to totally disconnected “repelling sets.” The grainy gray area is the interesting region. This is the “coastline”; it is itself a fractal object, revealing infinite complexity under magnification. Figures VIII.244 and VIII.245 show magnifications at two places on the coastline. The grainy areas revealed by magnification resemble pictures of the repelling set of the dynamical system at the corresponding values of λ .

4.3. This exercise refers to the family of dynamical systems $\{\mathbb{C}; z^2 - \lambda\}$. Use Algorithm 1 in section 4 with $-0.25 \leq \lambda_1 \leq 2$, $-1 \leq \lambda_2 \leq 1$, and $Q = (0.5, 0.5)$ to make a picture of the “Mandelbrot Set” $\mathcal{M}(0.5, 0.5)$. An example of such a set, for a different choice of Q , is shown in Figure VIII.246.

Figure VIII.243. A map of the family of dynamical systems described in example 4.2, computed using Algorithm 1 in section 4. The parameter space is $P = \{\lambda \in \mathbb{C}; 0 < \lambda_1 < 1, 0 < \lambda_2 < 1\}$. The gray grainy area is the interesting region. This is the “coastline”; it is itself a fractal object, revealing infinite complexity under magnification.

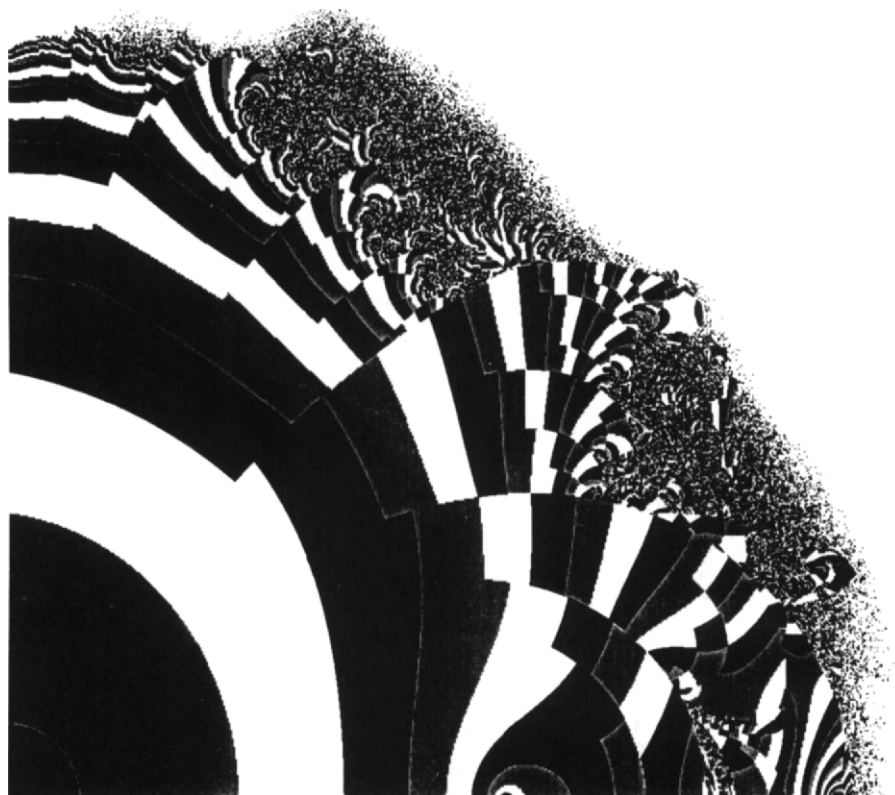


Figure VIII.244. Zoom on a small piece of the foggy area in Figure VIII.242. In it one finds grainy areas that resemble the repelling sets of the corresponding dynamical systems. At what value of λ does one find them? At the value of λ in the map where the picture you are interested in occurs.





Figure VIII.245. Zoom on a small piece of the foggy area in Figure VIII.242. The grainy areas in this picture here have different shapes from those in Figure VIII.244.

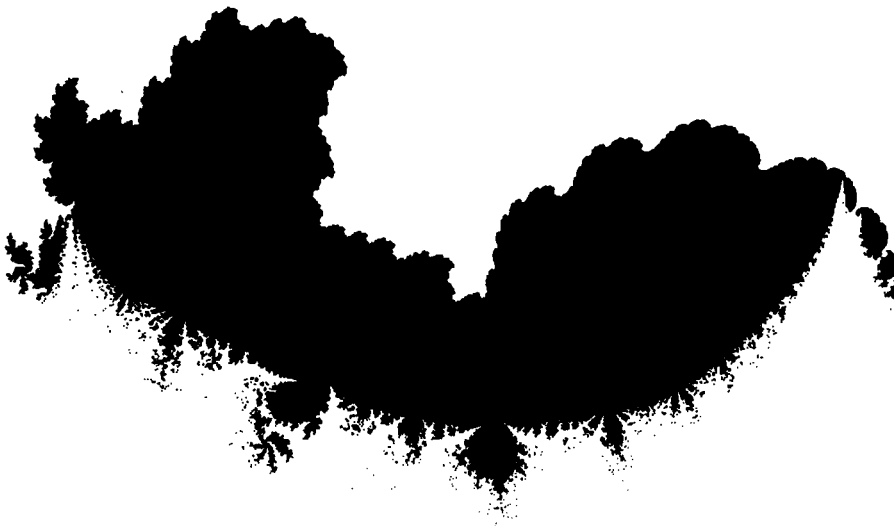


Figure VIII.246. A “Mandelbrot set” $\mathcal{M}(z_0)$ associated with the family of dynamical systems $\{\mathbb{C}; z^2 - \lambda\}$. This was computed using escape times of orbits of the point $z = z_0$ different from the critical point, $z = 0$.